Cosmic Quarry Mining Company Insurance Portfolio

The Pricing Frontier

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Executive Summary

Galaxy General Insurance Company (GGIC) is considering whether to expand its portfolio by responding to a request for proposals issued by Cosmic Quarry Mining Corporation (CQMC) for space mining insurance coverage. The opportunity is strategically important because a successful proposal would position GGIC as a leading insurer in interstellar commercial risk.

CQMC's operations currently span three solar systems: Helionis Cluster, Bayesia System, and Oryn Delta. The request for proposals asks for insurance solutions across four hazard areas: equipment failure, cargo loss, workers' compensation, and business interruption. This report evaluates whether GGIC should add products covering these four areas for CQMC, and if so, how these products should be designed.

Recommended Product Portfolio

We recommend offering products across all four hazard areas, subject to product-specific limits, exclusions, and pricing adjustments designed to reflect the distinct operating conditions across the three solar systems.

Equipment failure: Offer comprehensive cover with system-specific deductibles, maintenance-based eligibility conditions, and differentiated pricing for equipment mixes. With thousands of assets deployed across systems that are frequently exposed to a variety of harsh environmental and operational conditions, equipment failure represents a natural insurable hazard.

Cargo loss: Offer limited cover with a strict deductible, per-shipment policy limit, shipment caps, and targeted commodity exclusions. Cargo loss presents an attractive insurance opportunity only if the very severe outcomes related to large shipments of precious materials can be protected against.

Workers' compensation: Offer a master policy with system-specific endorsements reflecting differences in environmental hazards, communication delays, and injury severity. This hazard requires more special treatment in product design due to differences across solar systems and working areas in terms of operations and occupations.

Business interruption: Offer partial cover in a controlled form only, with carefully defined triggers, exclusions, and a benefit design that protects GGIC against highly skewed all-or-nothing loss behaviour. Although the underlying claims profile is difficult to model, a restricted product remains feasible where coverage is limited to well-defined business interruption events.

Strategic Rationale

The products are designed to remain profitable under base conditions and moderate stress scenarios. Pricing is conducted through an actuarial framework, focusing on expected loss, volatility, tail behaviour, and capital requirements, along with short and long-term considerations, in capital modelling. Equipment failure and Workers' compensation present

strong opportunities for insurance due to their relatively more straightforward exposure characteristics and claims behaviour, whereas cargo loss and business interruption require more constraints. Cargo loss exhibits extreme severity risk in certain cargo types and shipment volumes, and business interruption displays all-or-nothing claim behaviour in that many zeroes are present, but claims are severe when they occur.

Solar System Tailoring

The Oryn Delta system presents the most risk across all four hazards due to a host of environmental and operational hazards, including low visibility, asteroid ring hazards, and fluctuating gravity, which is reflected in a substantively larger tail risk. Comparatively, the other two systems are relatively stable and differ in riskiness depending on the hazard under consideration. The Bayesia system occasionally exhibits large spikes of electromagnetic and particle radiation, which are particularly detrimental to workers' compensation and equipment failure. The Helionis Cluster suffers from frequent communication blackouts due to asteroid clusters, which is more disadvantageous to the cargo loss and business interruption hazards. The products we suggest are inherently tailored to these differences, and in some extreme cases expert judgement is used to account for unmodelled risk factors.

Scalability and Future Adaptability

The proposed products are intended to be scalable as mining operations expand and as future hazards evolve. In particular, the pricing frameworks used for cargo, equipment failure, and business interruption can be updated as more experience data becomes available. The workers' compensation product is also designed to be reviewed periodically as CQMC's operational profile changes.

Scalability depends on regular repricing, ongoing data collection, and periodic review of exclusions, limits, and risk loadings. Future technological change, automation, route expansion, and new system conditions may present new challenges as they arise. It is important that GGIC obtains the most up-to-date data possible so that capital models can be updated dynamically.

Enterprise Risk Management Integration

The recommended portfolio should be presented to CQMC not only as a transfer of insurance risk, but also as part of a broader enterprise risk management (ERM) framework. Insurance can stabilise CQMC's cash flows and capital planning by converting uncertain large losses into known premium outflows. At the same time, the proposed products create incentives for better maintenance, safer work practices, disciplined cargo planning, and stronger operational controls. For that reason, the insurance program should be positioned as complementary to CQMC's internal controls rather than a substitute for them.

Product Design

Equipment Failure

GGIC should offer an equipment failure insurance product which is indemnity based and incorporates system-specific deductibles, rate adjustments, and Buhlmann credibility pricing.

Deductibles are based on the 10th percentile of historical claim amounts in each system, which helps lower total premiums and ensures exposures at lower risk of failure do not inflate gross premiums or capital requirements. These equate to values of Đ35,983 for Bayesian System, Đ31,537 for Helionis Cluster and Đ39,365 for Oryn Delta, reflecting the relative risk posed to equipment failure by each system. Any loss in excess of the deductible would be indemnified by GGIC.

This product provides comprehensive coverage for both idle and active equipment, including:

- Breakdown & Failure – Any accidental or unexpected breakdown of equipment;
- System wide events – system-specific events that impact equipment such as gravitational gradients, unpredictable flares, extreme temperatures, and fragmentation events;
- Galaxy wide events – losses due to galaxy wide shocks such as solar storms and extreme radiation;
- Environmental degradation – breakdowns due to degradation caused by long term exposure to radiation, debris, and variable temperature over time;

Exclusions will apply if the following conditions are not met:

- Equipment must be maintained according to the required maintenance standards (A.3 – Equipment Failure);
- Equipment must not be more than 20 years old;
- Equipment failure caused due to the mishandling of equipment will not be covered;
- Invalid equipment documentation regarding usage intensity, exposure period, maintenance and servicing will not be covered.

Cargo Loss

GGIC should offer a cargo loss product, but only with a tight risk-control design including a uniform deductible, a per-shipment policy limit, shipment caps, and commodity-specific restrictions. Without these controls, the severity profile appears too extreme for an affordable and sustainable product.

The deductible was set to be Đ100k to prevent attritional claims from inflating premiums. The policy limit was set at Đ1M per shipment to truncate the maximum payout, reducing the impact of very extreme losses. This still provides substantial coverage; 75% of the predicted claims fall within this limit.

This product provides coverage for any physical loss or damage arising during cargo shipment provided the loss falls within the range of deductible and policy limit.
Exclusions are carefully selected to control tail risk and uncertainty:

- cargo shipments carrying gold are excluded from cover;
- platinum is covered up to a 50,000kg weight cap;

- each insured vessel is subject to a maximum number of insured shipments by system. Helionis system vessels can be insured for their first 4 shipments, Bayesia vessels for their first 3, and Oryn Delta vessels for their first 2.
- Cargo shipments exceeding the recommended maximum cargo weight are excluded

Workers' Compensation

GGIC should offer a workers' compensation product structured as a single master policy with solar-system-specific endorsements reflecting the differing operating environments across the three systems.

Benefit schemes are constructed as a blend of standard industry practices, capital modelling and expert judgment. The structure of the master policy by solar system is summarised in Table 1, including stepped wage replacement, permanent disability benefits, and unique special benefits for particularly risky systems that modelled risk factors do not account for entirely. Detailed rationale for the specific structuring is contained in the (A.1 – Workers Compensation). Coverage as standard indemnities will also be available for certified medical facilities and for psychological rehabilitation.

Feature	Helionis Cluster	Bayesia System	Oryn Delta
Waiting period	7 days	None	None
Wage replacement steps	100 → 80 → 60%	100 → 85 → 70%	100 → 85 → 65%
Permanent disability	Max(5× salary, Đ250K)	Max(5× salary, Đ250K)	Max(5× salary, Đ250K)
Special benefit	—	Radiation illness: 3× salary (30-day latency)	Asteroid ring hazard: 3× salary

Table 1: Benefit structures for Workers' Compensation Product.

The following situations constitute exclusions from endorsements across systems:

- Off-duty recreational activities or personal travel resulting not required by the employer resulting in injury;
- Injuries from deliberate failure to use mandatory PPE where equipment was available and operational;
- Pre-existing conditions not materially aggravated by occupational exposure;
- Intentional self-injury, including prohibited substance use where this is a material cause;
- Bayesia System only: radiation illness from non-occupational sources.
- Claims not reported within 180 days of incident (Oryn Delta: clock starts from communications restoration).

Business Interruption

GGIC should only offer business interruption cover in a carefully constrained form due to a highly skewed and concentrated claims structure. We propose an indemnity-based coverage

that is sensitive to the claims clustering near the upper threshold of the policy limit through coverage triggers and exclusions.

This product implements a Đ28,000 deductible for each claim to ensure that any low-impact operational disruptions remain with CQMC rather than being transferred to GGIC. It also constitutes a Đ1,426,000 policy limit to protect against extreme loss outcomes.

Along with the deductible and policy limit, the following coverage triggers apply:

- A failure or required emergency overhaul of a core mining system
- System failure occurring while the unit is operating within its mandated service and maintenance window;
- Documented interruptions in the Quantum-secured supply chain for critical mineral extraction in the outer systems

Outside of these situations, business interruption insurance should not be offered. Exclusions apply in the following situations:

- A loss occurs when maintenance falls below the required benchmark;
- Interruptions result from poor maintenance or neglected infrastructure;
- Penalties, fines, and settlements arise from regulatory, environmental, or legal actions from the Interstellar Court of Environmental Justice

Pricing and Capital Modelling

The proposed products were assessed using a mix of frequency-severity modelling, credibility adjustments, and Monte Carlo simulation. Across the hazards, modelling approaches aimed to estimate loss distributions, which we summarise through expected costs, variances, and tail-related measures including Value at Risk (VaR) and Expected Shortfall (ES). We adopt a frequency-severity approach to modelling claims in each hazard through Generalised Linear Models (GLMs). Technical details of each modelling approach are provided in the (A.3).

Table 2 summarises the features of the loss distributions in the short term (1 year), aggregated over solar systems, for each hazard. It includes the annual premiums according to the product schemes outlined above, as well as net revenues and loss ratios which are calculated based on expected losses. Since premiums are fixed, returns have the same variance as losses. Table 3 provides the same metrics in the long term (10 years). Tables 4 and 5 summarise stress-testing results and report the impact on profits in the context of these extreme loses being realised.

Metric	Equipment Failure (ĐM)	Cargo Loss (ĐM)	Workers Comp (ĐM)	Business Interruption (ĐM)
Premium	44.61	660.1	17.01	206.25

Expected Loss	26.7	519.6	8.82	160.88
Expected Net Revenue	15.88	107.5	8.19	46.37
Std. deviation of loss/return	10.38	19.37	0.786	7.87
Expected Loss Ratio	59.9%	78.7%	51.8%	78%

Table 2: Short term (annual) features of costs, returns, and net revenue

Metric	Equipment Failure (ĐM)	Cargo Loss (ĐM)	Workers Comp (ĐM)	Business Interruption (ĐM)
Premium	436.8	6,337	189.5	2065.0
Expected Loss	268.0	5194	98.4	1608.8
Expected Net Revenue	168.8	107.5	91.1	453.7
Std. deviation of loss/return	33.4	61.0	2.62	24.9
Expected Loss Ratio	61.4%	82.0%	51.9%	77.9%

Table 3: Long term (10 yr) features of costs, returns, and net revenue

Metric	Equipment Failure (ĐM)	Cargo Loss (ĐM)	Workers Comp (ĐM)	Business Interruption (ĐM)
99% VaR loss	33.53	565.5	11.74	176.31
99% ES loss	129.1	573.1	12.57	178.58
Expected Net Revenue under VaR	11.08	61.5	5.26	29.94
Expected Net Revenue under ES	-83.49	53.9	4.44	27.67

Table 4: Short term (annual) stress testing results and impact on net revenue

Metric	Equipment Failure (ĐM)	Cargo Loss (ĐM)	Workers Comp (ĐM)	Business Interruption (ĐM)
99% VaR loss	371.5	5,334	104.5	1657
99% ES loss	420.9	5,357	105.4	1664
Expected Net Revenue under VaR	65.3	1,003	85.0	408
Expected Net Revenue under ES	15.9	980.3	84.1	401

Table 5: Long term (10 yr) stress testing results and impact on net revenue

Risk Assessment

Key Risk Identification

CQMC faces a wide range of risks that differ between solar systems. Each hazard area is affected by certain risks to varying degrees. The risk identification table outlines the potential risks associated with each solar system and hazard which were considered during the modelling process of GGIC's pricing and capital requirements.

Risk	Helionis Cluster	Bayesia System	Oryn Delta	Hazards Affected

Solar radiation exposure	Low	High	Low	CL, EF, WC
Solar storm/debris/asteroid event	High	Medium	Low	CL, EF
Gravity-related injury	Low	High	Medium	WC
Communication Blackout	High	Low	High	BI, CL, WC
Aged vessels	High	Medium	Low	CL, EF
Training deficiency	Low	Medium	High	BI, EF, WC
Route disruption	High	Low	High	CL, WC
Incomplete historical relevance	Low	High	High	All
1-in-100 year shock event	High	High	High	All
Psychological stress injury	Medium	High	High	WC
Power system inadequacy	Low	High	Medium	BI
High Utilisation Stress	High	Medium	Low	CL, WC

Table 5: Risk Assessment

Correlated Risk Scenario's

Correlated risks can have a major impact on the severity and frequency of claims in each solar system and hazard area. 1-in-100 year Galaxy-wide solar storms are a key example of a cross-hazard risk that has a negative effect on claim frequency. Such events drive the tail losses and volatility of losses in each solar system. These effects are captured within the modelling for cargo loss and equipment failure wherein the extreme tail losses represent the potential need for higher capital. Furthermore, cascading cross-hazards need to be considered as a workers compensation injury claim can be the result of the unexpected breakdown of equipment (A.4 – RCD Tool). Due to the limitations with data and assumptions of independence between hazards, such cases were not captured within the pricing and capital models. Consequently, there may be an underestimation of riskiness across solar systems and hazard areas leading to an understatement in premiums and capital. If it is observed that the capital models do not adequately hedge against correlated risk scenarios, we could potentially consider reinsurance schemes for hazard areas other than workers compensation to alleviate the financial pressure of high correlated risks. Additionally, constant re-assessment of modelling techniques and updating of data will be conducted to ensure the models can adapt and scale with the changes in the claim trends. Dynamic features have been included within the models to leverage new data that CQMC's operations will provide.

Scenario Testing:

Case	Equipment Failure (ĐM)	Cargo Loss (ĐM)	Workers Comp (ĐM)	Business Interruption (ĐM)
Best	29.0	519.6	8.82	16.28
Moderate	33.5	603.7	11.74	20.14
Worst	128.4	928.0	12.57	19.81

Table 6: Scenario testing annual losses

The scenario testing reveals the severity of high stress environments for each of the hazard areas. Particularly, the worst case for cargo loss and equipment failure both of which show

significant increases compared to moderate claim scenarios. This can be attributed to the direct exposure of vessels and equipment to 1-in-100-year events such as solar storms. Cargo loss experiences a rise in claims by 79% and equipment failure sees 4.4 times increase in losses due to the high joint shock factor in the extreme case (Table 6). The moderate cases are relatively less volatile in comparison, only increasing losses by 16% and 15% respectively. For both hazards, the extreme and moderate losses are driven by their frequency rather than claim size. Workers' compensation and business interruption claims tend to remain stable even in the worst case. They are not impacted by rare solar system events as potential risks have been captured by the correlated tail risks.

Threat Table:

Rank	Threat	Likelihood	Severity	Mitigation Technique
1	Galaxy-wide catastrophic shock	1% per year	Extreme	Policy limit, ES 99% required capital & reinsurance schemes
2	High value cargo loss	High	Extreme	Shipment caps & policy limits
3	Oryn Delta's environmental effects	Medium	High	Premium risk loading, shipment cap & deductible
4	Bayesian System radiation	Medium	High	Policy limit, shipment cap & deductible
5	Power system inadequeacy	Medium	High	Strict coverage triggers & policy limit
6	Quantum Bore failure	High	Moderate	Deductible, & premium weighting
7	Incomplete historical relevance	Inherent	Moderate	Bühlmann credibility weighting, judgement based analysis
8	Environmental & Operational injury	High	Moderate	Strict coverage triggers & system-specific modelling

Table 7: Shows threat, its likelihood and severity alongside mitigation techniques

Solar System Risk Narrative:

Based on the analysis of historical claims and the characteristics of the solar systems. Oryn Delta currently presents the highest risk to CQMC. Across all hazards, Oryn Delta's environmental risks have a significant impact on operations within the system. The low-visibility, unpredictable flares and hazardous zones create difficulties in traversal, safety and mining. This can be evidenced through the severity distribution of each hazard area which tends to have thicker tails compared to the Helionis Cluster and Bayesian System (A.3). Since the data is limited, and the Oryn Delta has only just been introduced to CQMC's mining operations, credibility weightings are applied in all hazards areas to account for the risk. For workers compensation and equipment failure, the Bayesian system is the 2nd riskiest with the Helionis Cluster being the least risky. The Bayesian System's radiation spikes and high

temperatures make equipment and mining operations difficult in the system. For Cargo loss and business interruption, the Helionis Cluster is the 2nd riskiest solar system. This difference can be attributed to the unique characteristics of each solar system and the operations of each hazard area wherein cargo loss and business operation experience greater losses with the communication blackouts and shifting debris.

Environmental, Social & Governance Considerations

To ensure sustainable operations within the galaxy, GGIC should consider integrating an ESG framework to support their policies. This can be achieved through implementing deductibles and strong maintenance conditions for cargo and equipment that CQMC will have to adhere to, reducing the potential risk of environmental damage through failure and cargo loss. Furthermore, the business interruption policy should include an exclusion for penalties and settlements from the Interstellar Court of Environmental Justice (ICEJ). For the social aspect, the workers' compensation product provides protection to workers from the risk hazards present in each solar system, ensuring they are adequately compensated. As for governance, GGIC must partake in active revaluation of pricing and capital, ensure transparency with CQMC and reflect accountability. Doing so will allow GGIC to maintain their compliance standards, financial stability and integrity.

Assumptions

Throughout the modelling process, several decisions and assumptions had to be made to build the insurance products for CQMC. These assumptions simplified the pricing and capital models, allowing them to be more robust for future modifications. However, the assumptions made need to be monitored as new data from CQMC and GGIC comes in, ensuring that the models are accurately portraying risks and trends. There are common assumptions, and some unique to hazard areas as seen in the Table 8.

Assumption	Rationale	Hazard Affected
Stationary & independence of severity & frequency distributions	Provides a stable baseline for long-term modelling despite not having yearly data	All
Constant inflation rate	Assuming an average inflation rate for long-term modelling due to the lack of data	All
Linear business growth	The 25% and 15% 10 year growth has no specific data to justify a non-linear growth pattern	All
Workforce allocation by mine count	Workforce allocated based on historical mine data due to a lack granularity in personnel data	WC
Fixed permanent disability benefit floor	A floor was of Đ250,000 was assumed to ensure equity	WC

Constant shock factor/probability	Shocks impact solar systems equally due to a lack of extreme event data	EF
Constant credibility weighting	A constant weighting decided based on judgement due to the lack of new data	EF
Policy constraints are strictly adhered to	Simplifies the modelling where exclusions are not accounted for in the model	CL
Shipment frequency parameters align with maximum shipments	Ensures expected shipments is equal to the number needed to meet each of the solar systems	CL
Stationarity of zero-inflation parameter	Stationarity is used due to a lack of current and relevant data	BI
1-to-1 relationship between number of units and total risk	Assumed based on systemic independence of risk and units	BI

Table 8: Assumptions, alongside the rationale behind choosing them and subsequent hazards affected

Data & Data Limitations

The data for the models were sourced by GGIC and CQMC alongside interest and inflation data (SoA, 2026). Additionally, the model for workers compensation incorporated real world data to support pricing and capital (Dimson, Marsh & Staunton, 2024). Despite having access to a variety of datasets, many of them lacked the full quality that was required and limited the capabilities of the pricing and capital models. In the case that complete high quality data is provided in the future, the pricing and capital models will be updated accordingly.

Limitation	Impact	Severity	Mitigation
No claim data for Bayesian System or Oryn Delta	Requires using Zeta and Epsilon as proxy solar systems	High	Bühlmann Credibility, Monte-Carlo simulation
Historical range deviations	Useable data size was reduced by 10%	Low	Data cleaning & Exclusion of rows
Lack of time-based data	Unable to determine time-based trends in claims	High	Assumed stationarity and annual reviews to be conducted
Extreme claim severities in cargo loss	Skewed models heavily to the tail end	High	Inclusion of a Policy limit
Qualitative bias	Lack of data required professional judgement to evaluate certain model aspects	Moderate	Stress testing & Covariate adjustment
Missing data variables/levels	Unable to capture significant variables in modelling	Low	Binomial simulation and proxy adjustments

Table 9: Data limitations, their respective impact, severity and mitigation strategies

Appendix

A.1 Product Design Rational

Equipment failure:
For equipment failure, GGIC will be covering CQMC's active and idle equipment.

Benefit Structure
The insurance product follows an indemnity-based model which incorporates deductibles, rate adjustments and Buhlmann credibility pricing. For the deductible, the insured will be responsible for paying the set amount based on the solar system. Due to the higher risk in the Oryn Delta and the equipment being far newer (A.3), this system will pay a higher deductible compared to the systems. Any losses after the deductible is paid will be covered by Galaxy general insurance. The deductibles are based on the 10th percentile of historical claim amounts. This helps lower the total premium and ensures low risk failure in idle equipment or active equipment do not inflate gross premiums or capital requirements.

Solar System	Initial Equipment	Deductibles	Premium (ST)	Premium (LT)
Helionis Cluster	2,580	31536.8	**27,041,794**	**266,375,316**
Bayesia System	1,290	35983.4	**10,105,283**	**99,547,039**
Oryn Delta	860	39365.0	**7,461,801**	**70,825,158**
Total Portfolio	**4,730**		**Đ44.61M**	**Đ436.75M**

Cargo loss
Benefit structure
A Đ100k deductible and Đ1M policy limit per shipment was uniformly applied across all 3 solar systems. This structure was critical to removing the frequent, low-severity claims which accounted for approximately 20% of historical claims which would otherwise inflate the premium. The policy limit truncates the maximum claim payout, reducing the impact of extreme losses given the highly skewed long right tail distribution of claim severity. Despite this, the limit provides substantial coverage with 75% of the predicted claims according to the modelling falling below the policy limit for full coverage benefit (excluding the deductible), thus offering utility to CSMQ by insuring a large portion of the claim mix.

Coverage triggers
Coverage applies to physical loss or damage arising during cargo shipment provided the shipment remains within the same insured solar system.

Exclusions
Given the limitations of the exact nature of CSMQ's shipment plans and mining expectations, carefully selected exclusions were necessary.

- Cargo shipments carrying gold is excluded from coverage. Modelling results demonstrate a disproportionate increase in expected claim severity, 192 times high

than the baseline cargo type cobalt and 4 times larger than platinum. Introducing gold in the coverage given the unavailability of the gold extraction forecast created too much pricing uncertainty

- Platinum is covered subject to a 50,000kg weight cap. Given platinum is already associated with 48 times increase in expected severity, restricting the weight which had a statistically significant exponential relationship with claim severity helped to control the exposure
- Shipment caps were imposed for each vessel. Specifically, vessels in the Helionis system were insured for their first 4 shipments, shipments in the Bayesia ship were insured for 3 shipments and oryn fleet vessels were insured for 2 shipments. The limits were informed carefully so that CSMQ can still meet their expansionary mining goals with their available fleet whilst mitigating risk associated with CSMQ selectively insuring high risk shipments which would invalidate modelling assumptions used (see cargo loss figure 1 for shipment cap derivation)
- Cargo shipments exceeding the recommended maximum cargo weight are excluded from coverage (see cargo loss figure 2 for max weight of container vessels)

Workers' compensation

Galaxy General recommends a single master WC policy with three system-specific endorsements, reflecting the fundamentally different operating environments across the three solar systems. For Helionis Cluster, which has the most established operations and the longest claims history of the three systems, the product follows standard industry practice including a seven-day waiting period consistent with the NCCI workers' compensation framework, and a three-tier wage step-down from 100% to 60% over 90 days which balances income security against the moral hazard of indefinite full replacement. Bayesia System and Oryn Delta carry no waiting period because structural communication delays and uncertain medical evacuation timelines mean a waiting period would routinely deny legitimate claims that simply cannot be initiated on time and is a design choice grounded in the same rationale applied to remote mining operations under Australian and Canadian offshore WC frameworks (WorkSafe Queensland, 2022). Two special benefits address hazards unique to each frontier system. For Bayesia, a radiation illness benefit of three times annual salary covers conditions arising from the intense electromagnetic radiation generated during binary star alignment events, with a 30-day delay window to capture cases that emerge over time. For Oryn Delta, an asteroid ring hazard benefit of three times annual salary applies to injuries sustained during extraction operations within the asymmetric asteroid ring, where localised orbital shear and fluctuating gravitational gradients create materially elevated injury risk relative to standard surface mining. The Đ250,000 permanent disability floor addresses a structural inequity in salary-linked benefit formulas. At five times annual salary, a galley worker earning Đ 40,000 would receive only Đ 200,000 for total and permanent disability, while an executive earning Đ 500,000 would receive Đ 2.5M for the same injury. The floor ensures every worker receives a meaningful minimum benefit regardless of pay grade. Full benefits structure, coverage triggers and exclusions are provided in below.

Feature	Helionis Cluster	Bayesia System	Oryn System
Waiting Period	7 days(standard)	None(day-1 coverage)	None(day-1 coverage)
Indemnity - Phase 1	100% salary, days 1-7	100% salary, days 0-30	100% salary, days 0-14
Indemnity - Phase 2	80% salary, days 8-90	85% salary, days 31-180	85% salary, days 15-90
Indemnity - Phase 3	60% salary, days 91+	70% salary, days 181+	65% salary, day 91+
Medical coverage	Uncapped, certified facilities	Uncapped, certified facilities	Uncapped, certified facilities
Permanent disability benefit	Max(5 x annual salary, Đ250,000)	Max(5 x annual salary, Đ250,000)	Max(5 x annual salary, Đ250,000)
Special Benefit	-	Radiation illness: 3 x annual salary	Asteroid ring uplift: +20%
Psychological rehabilitation	Standard Indemnity	Standard Indemnity	Standard Indemnity

The following are excluded across all system endorsements:
• Off-duty recreational activities or personal travel not required by the employer.
• Injuries from deliberate failure to use mandatory PPE where equipment was available and operational.
• Pre-existing conditions not materially aggravated by occupational exposure (aggravation remains covered).
• Intentional self-injury, prohibited substance use where this is a material cause

Business interruption

Claims exhibit an all-or-nothing nature. Claims for the majority would be zero. If otherwise, there is a high likelihood of hitting upper threshold claim amount. Two benefit structures are explored:

- Option 1: an indemnity model using Gamma distribution with variable payout based on actual business interruption loss, with a 28000 deductible and 1426000 policy limit. This is appropriate specifically for attritional claims that are not at the 1426000 limit. The company then remains competitive and fair for premium charged.

- Option 2: a layered/tiered model with fixed/binary payout is efficient and economically viable for the majority of clients. Given a strong concentrated skew toward the maximum claim amount, a “total loss” tier would cover the most common amount i.e., the policy pays out 1426000 in full in case of interruption. This significantly reduces claims handling expenses and deem Galaxy General much profitable, as the company does not need to perform separate quantitative assessment of the amount. However this requires the cost of over-indemnifying to be compensated by reduced audit cost. Therefore, the company if implementing this option, would need stricter coverage triggers.

Coverage triggers are designed based on the significant covariates consistently found in the Zero-Inflated Negative Binomial, Negative Binomial and naive Poisson model for claim frequency:

- A claim is triggered only when operational loss exceeds 28000 per mining unit to ensure the policy ignores low-impact high-frequency attritional claims
- As payouts are highly sensitive to energy backup failures, any failure or required emergency overhaul of an AI-driven neuro-algorithmic mining system constitutes an automatic trigger
- An operational interruption is triggered if a system failure occurs while the unit is operating within its mandated service window
- Also, a trigger through documented interruptions in the Quantum-secured supply chain for critical mineral extraction in the outer systems

To further incorporate Option 2, a total failure is defined as a binary state where:

- The estimated downtime and recovery costs are projected to meet or exceed the upper threshold claim amount of 1426000.
- This trigger is primarily reserved for catastrophic events such as cluster-level fragmentation in the HC or total habitat depressurization in the Zeta or Epsilon systems.

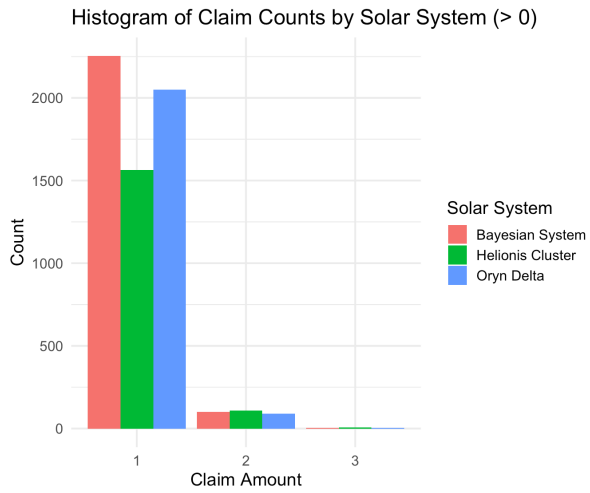
Exclusions:

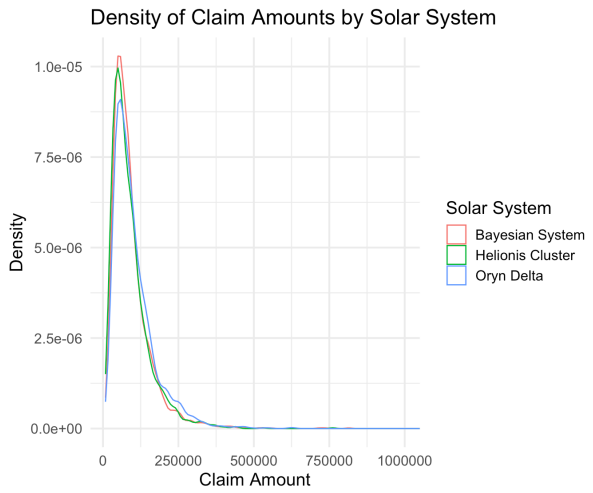
- Excludes losses where maintenance_freq falls below a benchmark of 4. Coverage requires verifiable telemetry from AI-driven neuro-algorithmic systems to ensure the $1,426,000 limit is not used to subsidise neglected infrastructure in hazardous environments
- Excludes all penalties and settlements from the Interstellar Court of Environmental Justice, including triggers for forward contamination, landscape alteration, and orbital saturation. The social and governance risk removal is supported by historical loss patterns, such as the 75 million fine in 2165 and 40 million settlements in 2174

A.2 Exploratory Data Analysis

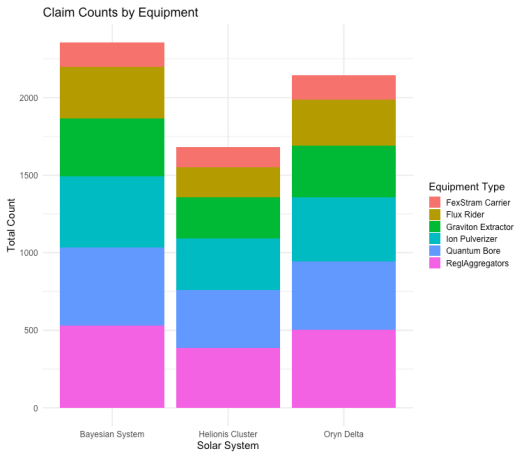
Equipment failure

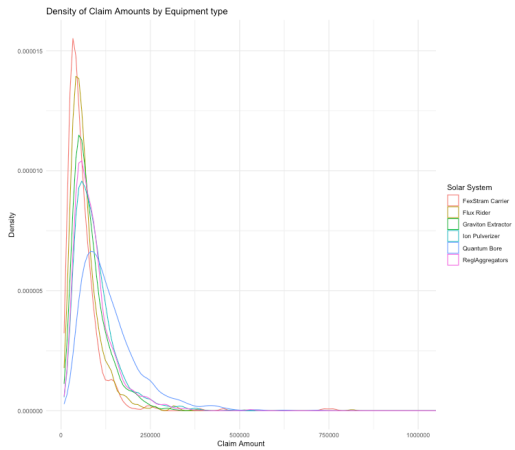
GGIC is offering CQMC an insurance product to provide protection for the costs associated with equipment failure. Due to the nature of CQMC's business operations, a significant amount of their equipment related assets are exposed to harsh planetary environments and unexpected conditions. With a total of 4730 assets, the insurance product will safeguard CQMC from risks that may be a detriment to their growth and plans for expansion.
The summary statistics of historical claims indicates high risk in the Zeta system (Oryn Delta proxy) due to it having thick tails and being heavily right skewed as seen by its kurtosis of 404 and skewness of 16.3 (Claim Severity Table). The Epsilon System, proxy for the Bayesian system, has a kurtosis of 149, demonstrating high risk in its tails compared Helionis Cluster with a kurtosis of 97. Zeta's expected shortfall (ES) of Đ419,203 illustrates the significance of tail losses and the need to address it in the insurance product. Typically, most equipment experience no claims, however, historical incidents indicate that 1 claim per year is most likely with multiple claims on the same equipment being significantly less likely. Due to the high risk associated with Zeta, it has the highest frequency for 2-3 claims (Claim Frequency Table). The ranking of risk between solar systems can be further corroborated by the average risk index wherein the equipment in Oryn Delta has an average risk index of 0.62 (Equipment Risk Table) compared to indices of 0.53 and 0.49 for the Bayesian System and Helionis Cluster respectively.





Quantum Bores present the greatest risk across all solar systems, having the highest average risk index of 0.8. They also have the highest mean claim of Đ142,623.36 and expected shortfall of Đ495,958.70. This can be attributed to being exposed to harsh environments and having to drill through tough materials that accelerate degradation and breakdowns. Despite having the second highest average risk index of 0.72, FexStram Carriers have the lowest mean claim amount and also a significantly lower tail risk.

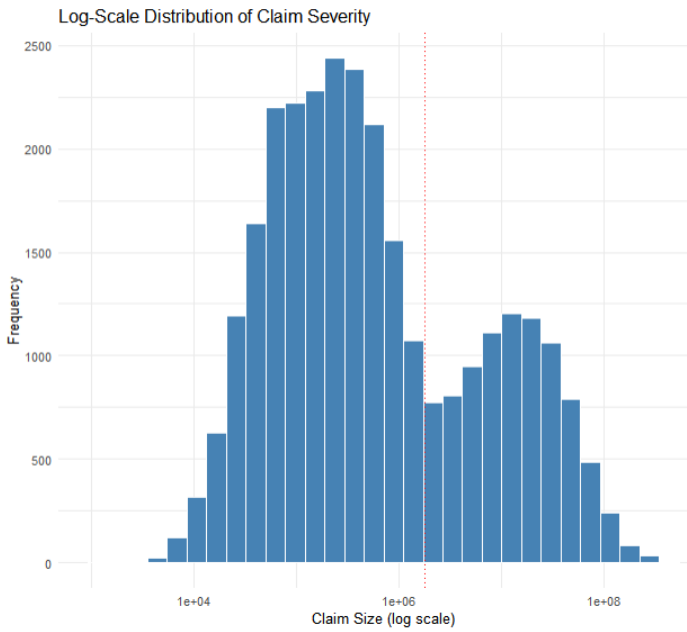




Cargo loss

GGIC is recommending offering a cargo loss insurance product for the Helionis, Bayesia and Oryn system cargo fleet. Our proposed product has a Đ660M annual premium and is expected to generate a Đ107M return for GGIC. Our product's loss ratio of 78.7% is slightly higher than industry standards (average cargo insurance loss ratio was 70.2% in Europe from 2015-2025) which is however warranted given the major frequency and severity risks associate with interstellar cargo loss. Our historical data suggests that 18.2% of shipments result in at least one claim. This is coupled with the high claim severity, with the average and median claim loss being Đ7.8M and Đ382k respectively. Furthermore, the histogram reveals a highly skewed bimodal distribution, with a few observations with substantially high claim sizes of over Đ680M. Full coverage of cargo loss would accrue over Đ10 billion, requiring policy designs and restrictions to minimise risk and develop an affordable premium for CQMC.

Our products design features include a 100k deductible to cut out 20% of the frequent but small claims that arise and a Đ1M policy limit to truncate claim size severity from large unbounded losses. Our policy design also includes annual shipment caps tailored to the mining output needs of each solar system. Specifically, Helionis, Bayesia and Oryn are capped to 4,3 and 2 shipments respectively to hedge exposure. Finally, we also have a gold cargo type exclusion and a restriction of platinum to 50,000kg per shipment to mitigate against high-risk shipments, especially in the Oryn system that contains dense deposits of rare metals. The exclusion of gold was especially necessary given 89% of the 5% largest claim sizes were attributed with gold cargo types.



Our product is both adaptable and scalable for evolving future risks. Our pricing framework is simulation driven, meaning changes to the environment or unforeseen expansionary plans can be reflected as adjusted inputs to redesign premiums or policy limits without changing the model structure. Our current product was developed to be adaptable against changes in future risk as shown in our moderate stress test scenario which still achieved a positive return of Đ23M on average even under these strained conditions.

Our product plays an important role in the CQMC's enterprise risk management. It transfers their risk of infrequent but large claims to an insurer for a fixed premium. This protects CQMC's risk exposure against adverse scenarios. In return for a fixed premium, CQMC's develops a more predictable financial forecast, which assists when it comes to budgeting for future projects and investments.

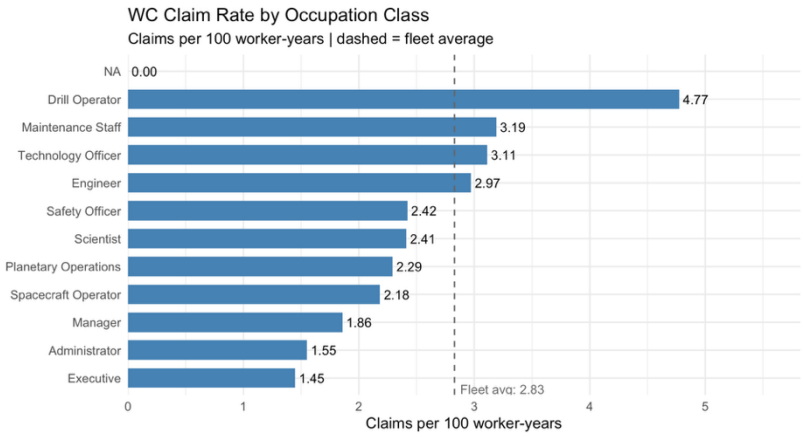
Workers' compensation

Cosmic Quarry Mining Corporation operates 55 active mines across three solar systems, with a workforce of 35,809 employees distributed across systems in proportion to mine count: Helionis Cluster has 30 mines with 19,532 workers, Bayesia System has 15 mines with 9,766 workers) Oryn Delta has 10 mines with 6,509 workers as shown in Figure 1A. Each system presents a distinct physical environment that directly shapes the worker's compensation risk. Helionis Cluster orbits a stable G2V star and operates across two geologically quiet planets making it the lowest-risk system of the three. However, two large asteroid clusters in the outer system produced fragmentation events that periodically require satellite repositioning, creating communication blackouts that delay injury reporting and slow claims processing. This effect is observable in the data with Helionis recording a mean claim duration of 64.5 days which is 11 percent above the fleet average, despite being the safest system by claim rate. Bayesia System orbits a tight binary star pair and the key worker's compensation concern is that at certain orbital alignments, the two stars combine to produce sharp spikes of electromagnetic and particle radiation extending through the inner system where CQ

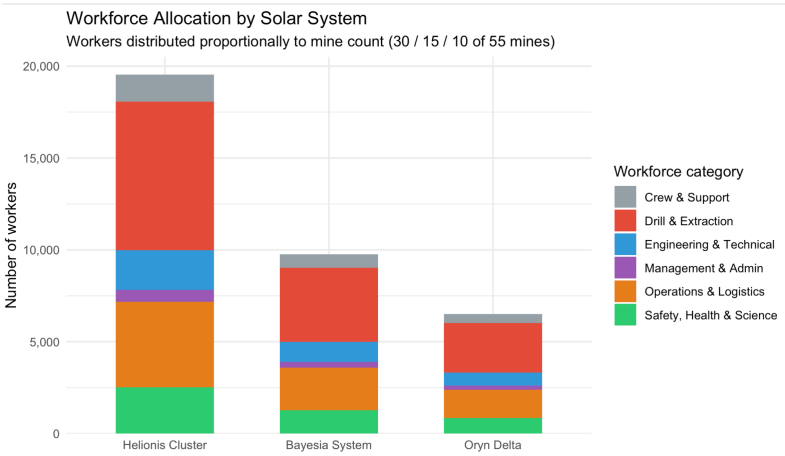
operates. The planet has a thin magnetosphere which provides limited shielding, leaving surface workers exposed to elevated radiation between alignment events as well as acute spikes during them, while the planet also has high gravity which places significant biomechanical load on workers across all physical roles. Because Bayesia has no historical claims data, pricing relies on Zeta as the closest gravity proxy. Oryn Delta orbits a dim M3V red dwarf and the low luminosity of this star produces a permanently high-contrast and low-visibility environment across the entire system with workers facing elevated navigation and proximity accident risk as a structural feature of their daily operations. CQ's active expansion into the systems asymmetric asteroid ring introduces fluctuating gravitational gradient hazards with the exoplanet beyond the ring producing rugged tectonic terrain introducing seismic injury exposure. Long-range communication infrastructure remains inconsistent which delays medical evacuation and claim initiation which is analogous to but more persistent than the satellite blackout problem at Helionis. Because none of these specific hazards appear in the historical data, an expert judgement loading is applied on top of the base rate to reflect these unmodelled risk factors and an annual risk review is built into the product to reprice the system as operational data accumulates.

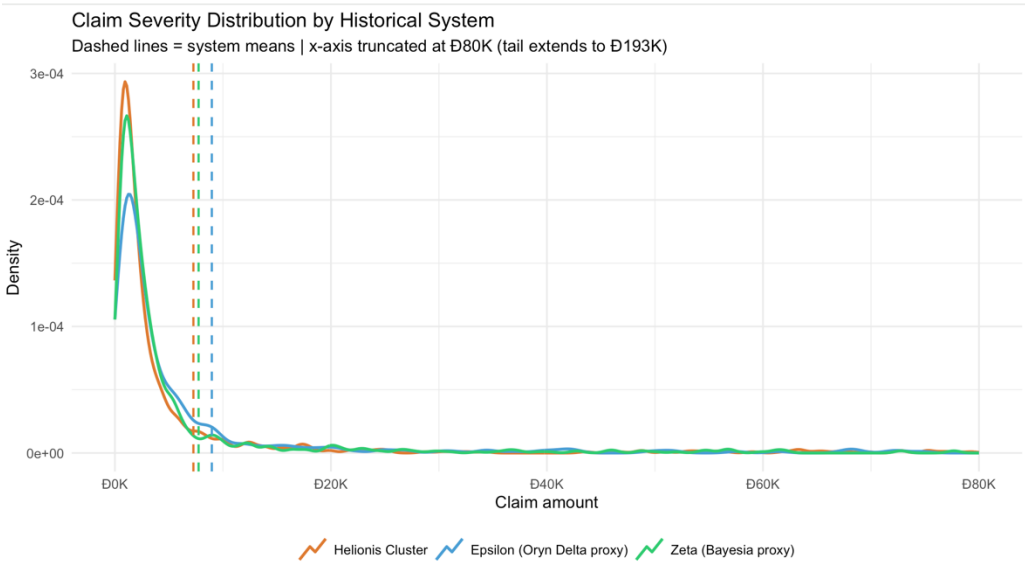
The workers' compensation claims database underpinning this pricing exercise contains 133,398 worker-period records and 1,913 paid claims across the three historical systems, representing 67,875 worker-years of total exposure with appendix Figure A2 showcasing the full frequency and severity distributions from this dataset. As seen in Figure , claim frequency varies greatly by occupation class with drill operators producing the highest rate at 4.75 claims per 100 worker-years, which is 69 percent above the fleet average of 2.81 and reflects their sustained exposure to heavy machinery and physically demanding operations. On the other hand administrators sit at the opposite end of the spectrum at 1.54 per 100 worker-years while CQ's largest occupation class of Maintenance Staff (34.3 percent of the total workforce) records a rate of 3.17 per 100 worker-years, meaning this group drives a disproportionate share of the aggregate expected claims simply through its scale.
Beyond occupation, four individual worker characteristics show clear and statistically significant relationships with claim frequency. Firstly workers with 20 or more years of experience produce 43 percent fewer claims than those with 5 to 10 years of experience, reflecting the potential importance of accumulated skill and hazard awareness. Secondly, a one-point improvement in safety training quality score reduces claim frequency by approximately 31 percent across the full range from score 1 to score 5, while a one-point increase in psychosocial stress score raises frequency by approximately 37 percent from score 1 to score 4. Finally, workers who have a prior accident record generate 28 percent more claims than those who do not and each of these factors are formally incorporated into the schedule modification factor described in the premium computation section.

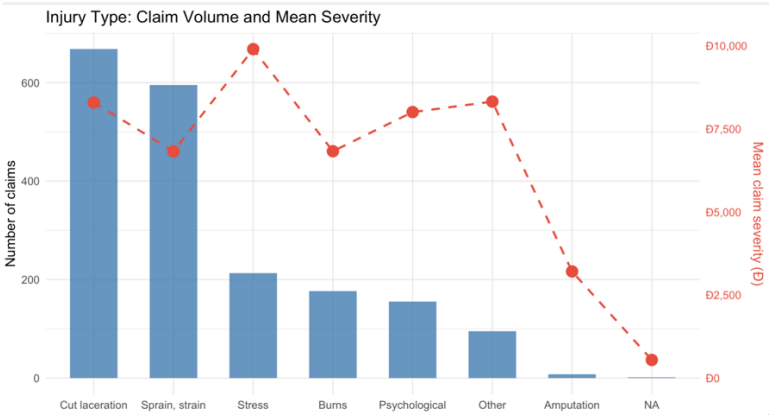
Turning to claim costs, severity is highly right skewed across all systems with the fleet mean of Đ7,823 per claim being 3.2 times the median of Đ2,054 while the largest single observed claim reaching Đ193,357. This skewness, visible in the density curves confirms that aggregate annual losses are disproportionately driven by a small number of very large claims rather than by the typical claim. The Lognormal distribution was selected over Gamma for

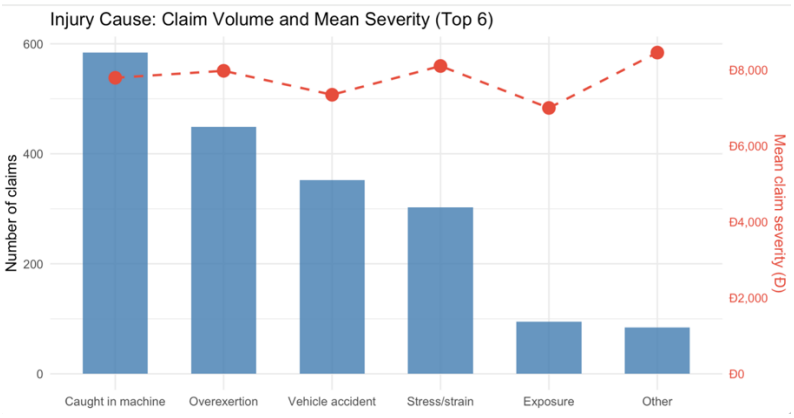


severity modelling at all three systems on the basis of AIC, with margins of 218, 431, and 467 points respectively, because it better captures this extended right tail. By injury type, stress injuries represent only 11.1 percent of all claims but carry the highest mean cost at Đ9,891, while cut and laceration injuries are the most frequent at 34.9 percent. The gravity impact is strongest when examining injury severity by system rather than by type as shown in with musculoskeletal injury severity rising from Đ5,615 at Epsilon (0.875g) to Đ10,615 at Zeta (1.375g) as gravity increases, confirming that Bayesia System carries not only higher claim frequency but also substantially higher per-claim costs than either Helionis or Oryn Delta.









A.3 Summary of Pricing and Capital Modelling

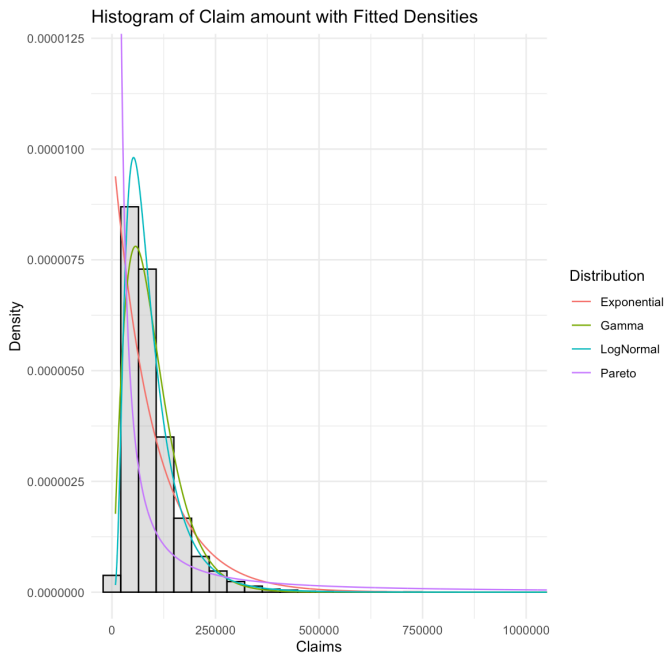
Equipment Failure:

In order to set premium prices that are catered towards the risk profiles of each solar system, a LASSO GLM was trained on a subset of the historical data. The severity model follows a gamma distribution with a log link function and the frequency model follows a Poisson distribution with a log offset to account for the exposure of the equipment. These models were selected due to their ability to capture the risks of each predictor. Other models such as random forests and base GLM's were tested though lacked interpretability or predicting

power. Upon predicting the severity and frequency, the expected loss was calculated by solar system and an empirical deductible factor was applied based on this. For each of the solar systems, a Bühlmann credibility weighting was applied to compensate for not having data from Oryn Delta and the Bayesian System. With the pure premium for each solar system, the annual gross premium for the short-term and long-term were calculated by factoring in equipment growth, mean equipment risk, fixed expenses of 500,000, a 7% variable expense ratio, 5% profit load which includes a 2% environmental load. The annual values were also adjusted for inflation and included the risk-free discounting.

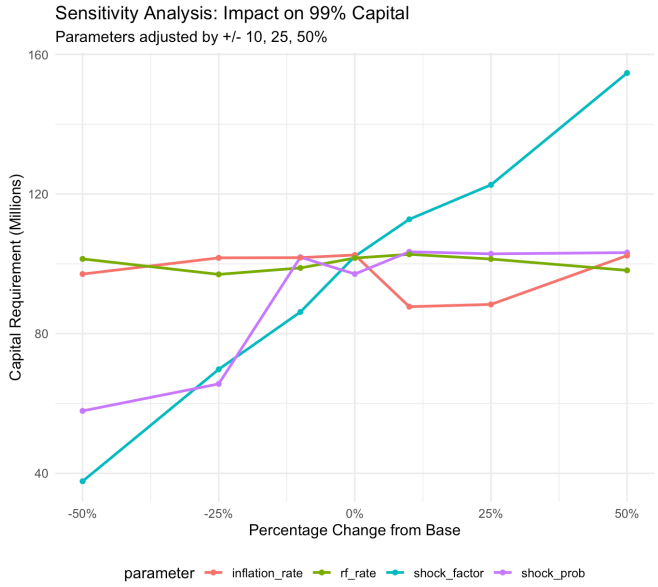
To model the capital, maximum likelihood estimation was employed to obtain theoretical distributions for each solar system, in all cases the log-normal and negative binomial distributions provided the best fit for the data based on the Akaike Information Criterion (AIC), Anderson-Darling tests and Kolmogorov-Smirnov tests. With these distributions, monte-carlo simulations were run, incorporating the deductibles and dependency structures between solar systems through joint shocks. The shock represents a 1-in-100 year event that multiplies the negative binomial success parameter by a factor of 5, increasing frequency of claims throughout the simulated year. The simulation randomly generated samples from the theoretical distributions which were summed and averaged at the end to provide an approximation of losses, and potential catastrophic years in the tails of the simulation.
The models were then used to predict the remaining severity and frequency in the data. This Monte Carlo Simulation – Capital Modelling which involves using maximum likelihood estimates to model distributions for the simulation For specific hazards where data wasn't enough, Buhlmann Credibility was applied to ensure the

Equipment failure – Modelled severity distributions



For equipment failure, the required capital is most sensitive to the joint shock factor and the probability of such a shock occurring between solar systems, impacting the standard deviation of yearly significantly. For every 25% increase or decrease in the shock factor,

there is an approximately 50% change in required capital showcasing the high volatility presented by this parameter which has a base value of 5. For the shock probability the base value is 0.01 suggesting a 1-in-100-year probability. The stress testing shows that it increases required capital until flatlines after going past the base rate, the trend in this case suggests that lowering the probability of a shock by 50% will half approximately half the required capital. Both factors showcase severe tail behaviour when modified. The randomness associated with the simulations shows small fluctuations for the inflation rate and interest rate however it is not substantial enough to have a major impact on the required capital.



Solar System	No. Equipment (ST)	No. Equipment (LT)	Premium (ST)	Premium (LT)
Helionis Cluster	2,645	3,225	**27,041,794**	**266,375,316**
Bayesia System	1,322	1,613	**10,105,283**	**99,547,039**
Oryn Delta	882	1,075	**7,461,801**	**70,825,158**
Total Portfolio	**4,849**	**5,913**	**Đ44.61M**	**Đ436.75M**

Claim Frequency Table

Solar System	Mean	Std Deviation	Total Claims Count
Bayesia System	0.072	0.270	2464
Helionis Cluster	0.105	0.330	1809
Oryn Delta	0.065	0.258	2242
Total			**6515**

Claim Severity Table

Solar System	Mean	Std Deviation	Skewness	Kurtosis	VaR (99%)	ES (99%)	Coef of var	Total Claim Amount
Bayesia System	93664	81042	8.35	149	211519	321907	0.87	241184382
Helionis Cluster	90662	73116	5.99	97	214453	299658	0.81	173798181

	1079					41920		
Oryn Delta	92	137931	16.30	404	247837	3	2.80	255616178
Total								**Đ670.60M**

Capital Modelling Table

Statistics	Short Term - 1 Year	Long Term - 10 Year
Expected Loss	26,729,750	268,006,956
Std Dev	10,383,466	33,400,055
VaR (95%)	29,032,833	360,378,538
VaR (99%)	33,527,164	371,512,908
VaR (99.5%)	128,455,196	379,471,176
ES (95%)	49,511,513	376,163,459
ES (99%)	128,143,331	420,857,313
ES (99.5%)	131,776,748	467,428,038
Capital (99%)	101,413,581	152,850,358

Equipment Risk Table

Equipment Type	Helionis Cluster	Bayesia System	Oryn Delta	Mean Risk
Quantum Bores	0.69	0.77	0.93	0.80
Graviton Extractors	0.48	0.57	0.73	0.59
Fexstram Carriers	0.67	0.71	0.78	0.72
ReglAggregators	0.24	0.26	0.35	0.28
Flux Riders	0.24	0.21	0.20	0.22
Ion Pulverizers	0.64	0.66	0.75	0.68
Mean	**0.49**	**0.53**	**0.62**	

Equipment Growth Table

Year	Helionis Cluster	Bayesian System	Oryn Delta
0	2580	1290	860
1	2644	1322	881
2	2709	1354	903
3	2773	1386	924
4	2838	1419	946
5	2902	1451	967
6	2967	1483	989
7	3031	1515	1010
8	3096	1548	1032
9	3160	1580	1053
10	3225	1612	1075

Helionis Severity Model Stats

Distribution	AIC	BIC	AD Stat	KS Stat	Parameter 1	Parameter 2
Exponential (λ)	47600.68	47606.24	129.3117	0.2023	0.00001103	-

Log Normal (μ, σ)	46736.82	46747.93	0.3907	0.0132	11.19671911	0.65271946
Gamma (ϒ, c)	46916.67	46927.79	11.3631	0.0542	2.44555477	0.00002697
Pareto (α, λ)	49716.74	49727.85	Inf	0.3616	0.46321405	8416

Helionis Frequency Model Stats

Distribution	AIC	BIC	Parameter 1	Parameter 2
Poisson (λ)	11970.78	11978.54	0.1045544	-
Negative Binomial (p, r)	11958.57	11974.08	2.3950656	0.9581719

Bayesian Severity Model Stats

Distribution	AIC	BIC	AD Stat	KS Stat	Parameter 1	Parameter 2
Exponential (λ)	64106.46	64112.31	207.9012	0.22306496	0.00001068	-
Log Normal (μ, σ)	62703.59	62715.3	1.3247	0.02098482	11.24430515	0.61350598
Gamma (ϒ, c)	63068.84	63080.54	Inf	0.06807492	2.61574804	0.00002793
Pareto (α, λ)	66591.59	66603.29	Inf	0.36530491	0.50392736	10508

Bayesian Severity Model Stats

Distribution	AIC	BIC	Parameter 1	Parameter 2
Poisson (λ)	18080.75	18089.2	0.07150527	-
Negative Binomial (p, r)	18077.69	18094.59	4.01489789	0.98250166

Oryn Delta Severity Model Stats

Distribution	AIC	BIC	AD Stat	KS Stat	Parameter 1	Parameter 2
Exponential (λ)	59602.16	59607.93	Inf	0.20971869	0.00000926	-
Log Normal (μ, σ)	58330.94	58342.48	1.753326	0.0190155	11.35082988	0.63832260
Gamma (ϒ, c)	58889.33	58900.87	Inf	0.07477441	2.24470356	0.00002079
Pareto (α, λ)	62160.26	62171.79	Inf	0.37971994	0.45889949	9621

Oryn Delta Frequency Model Stats

Distribution	AIC	BIC	Parameter 1	Parameter 2
Poisson (λ)	16864.83	16873.28	0.06521044	-
Negative Binomial (p, r)	16857.99	16874.88	2.71422729	0.97653826

Equipment Statistics

Equipment Type	Total Claims	Mean Claims	Std Dev Claims	Kurtosis	VaR	ES	Coefficient of Variation
Fexstram Carriers	31190281	61885.48	61940.83	66.20865	143956.5	239335.1	1.0008945
Flux Riders	63035269	68516.6	50336.17	58.43347	147312.9	217511.1	0.7346566
Graviton Extractor	88494443	83015.43	63248.7	126.43144	183621	252927.8	0.7618909
Ion Pulverizers	132664764	98124.83	93468.58	269.89012	215634.1	327997.9	0.9525478
Quantum Bore	209085850	142623.36	155240.69	360.22644	309570.8	495958.7	1.0884661

ReglAggregators	146128134	94215	89023	199	208752	318767	0.945

Severity LASSO Model (Bayesian System Baseline)

Predictor	Coefficient	Exponential coefficient	Risk Ranking
(Intercept)	10.82575349	50299.65458	1
Fexstram Carriers	-0.427266831	0.652289478	10
Flux Riders	-0.302030758	0.739315325	9
Graviton Extractor	-0.117750389	0.888917908	8
Ion Pulverizers	0.05813667	1.059859837	3
Quantum Bore	0.447782584	1.564838438	1
ReglAggregators	.	.	.
Equipment Age	0.015908963	1.016036184	5
Helionis Cluster	-0.050627161	0.950633037	7
Oryn Delta	0.123448498	1.131391734	2
maintenance_int	2.01785E-05	1.000020179	6
usage_int	0.026575055	1.026931321	4

Frequency LASSO Model (Bayesian System Baseline)

Predictor	Coefficient	Exponential coefficient	Risk Ranking
(Intercept)	-3.308555941	0.036568943	11
Fexstram Carriers	0.480455011	1.616809901	1
Flux Riders	0.288794295	1.334817122	3
Graviton Extractor	0.084615876	1.088298944	4
Ion Pulverizers	.	.	.
Quantum Bore	-0.106777177	0.898725907	9
ReglAggregators	-0.225100775	0.798435753	10
Equipment Age	0.057684575	1.059380788	5
Helionis Cluster	0.362548248	1.43698655	2
Oryn Delta	-0.085407456	0.918138108	8
maintenance_int	0.00023304	1.000233067	7
usage_int	0.041004073	1.041856349	6

Equipment Usage & Maintenance table

	Helionis Cluster		Bayesia System		Oryn Delta	
Equipment Type	% in Operation	Maintenance Schedule (Hrs)	% in Operation	Maintenance Schedule (Hrs)	% in Operation	Maintenance Schedule (Hrs)
Quantum Bores	0.95	750	0.80	600	0.75	500
Graviton Extractors	0.95	750	0.80	600	0.75	500
Fexstram Carriers	0.90	375	0.75	400	0.70	250

ReglAggregators	0.80	1,500	0.75	1,000	0.70	300
Flux Riders	0.80	1,500	0.80	1,000	0.75	300
Ion Pulverizers	0.50	1,000	0.60	750	0.50	500

Cargo loss:

Premiums were derived from Monte Carlo simulations derived from the glm distributional properties of claim severity and frequency. For severity, a 2 layered approach was used by segmenting claims into small and large claim size. This was necessary given the bimodal nature of claim severity which a singular distribution would struggle to capture. This was validated from a model assessment by splitting 80% of the data into a training set and testing on the other 20%. The 2 layered GLM model with a gamma distribution on statistically significant variables and splitting the claim size at 1.8M (equal to where the 2nd peak of the bimodal claim size begins) saw a 650k RMSE reduction when compared to a singular GLM. Although, the single severity model was still utilised for interpretation.

The premium composition started with the pure premium which was the expected loss of the simulations. Buhlmann credibility was used to stabilise the premium estimates by blending the container type estimates to the portfolio-level mean. Stress loading were applied to ensure premiums account for potential adverse scenario conditions. Additional risk loaded were added to reflect volatility associated with claim loss. Cost of capital and operational expense was added on too.

Solar system	Helionis	Bayesia	Oryn	Total
Vessels insured	1,160	1,128	774	3,062
Pure premium	286,026,916	152,013,507	81,563,628	519,604,051
Buhlmann credibiltiy premium	283,213,208	150,635,565	80,125,874	513,974,647
+Stress loading (12% worse 18% moderate)	309,720,579	174,939,198	91,074,367	575,734,144
-PV discounting of claims	307,504,089	173,687,260	90,422,601	571,613,950
+Volatility loading (150%)	329,054,526	189,506,772	101,908,464	620,469,762
+Cost of capital (7.46%)	332,004,766	191,612,802	103,445,634	627,063,202
+Operational expenses (5% of final premium)	349,478,701	201,697,686	108,890,141	660,066,528
Final premium	349,478,701	201,697,686	108,890,141	**660,066,528**
Premium per vessel	301,275	178,810	140,685	**215,567**
Loss ratio	81.8%	75.4%	74.9%	**78.7%**

Aggregate loss distribution

Short-term

Loss statistics	Helionis	Bayesia	Oryn	Total
Expected loss	286,026,916	152,013,508	81,563,628	519,604,052
Standard deviation of loss	14,366,958	10,546,341	7,657,242	19,372,444
VAR 95%	310,372,683	169,706,682	94,266,863	551,571,812
VAR 99%	321,067,044	177,930,554	99,969,086	565,539,346
VAR 99.5%	325,574,364	180,244,480	102,169,130	571,352,510
ES 95%	316,947,526	174,635,127	97,939,490	560,534,130
ES 99%	326,643,111	181,504,717	103,323,869	573,120,410

Return statistics	Helionis	Bayesia	Oryn	Total
Expected return	45,977,850	39,599,294	21,882,006	107,459,150
Standard deviation of return	14,366,958	10,546,341	7,657,242	19,372,444
VAR 95%	21,632,083	21,906,120	9,178,771	75,491,390
VAR 99%	10,937,722	13,682,248	3,476,548	61,523,856
VAR 99.5%	6,430,402	11,368,322	1,276,504	55,710,692
ES 95%	15,057,240	16,977,675	5,506,144	66,529,072
ES 99%	5,361,655	10,108,085	121,765	53,942,792

Long-term

Loss statistics	Helionis	Bayesia	Oryn	Total
Expected loss	2,859,054,809	1,518,792,352	816,230,628	5,194,077,789
Standard deviation of loss	46,085,415	32,726,617	24,248,874	60,953,173
VAR 95%	2,934,398,728	1,572,173,215	856,641,892	5,293,425,359
VAR 99%	2,967,871,509	1,596,979,199	873,375,078	5,333,802,728
VAR 99.5%	2,979,027,197	1,603,767,659	880,473,934	5,353,032,281
ES 95%	2,954,817,984	1,586,702,656	867,024,739	5,318,520,707
ES 99%	2,982,070,474	1,607,648,151	882,491,083	5,356,790,113

Return statistics	Helionis	Bayesia	Oryn	Total
Expected return	517,984,381	391,715,814	233,302,193	1,143,002,387
Standard deviation of loss	46,085,415	32,726,617	24,248,874	60,953,173
VAR 95%	442,640,462	338,334,951	192,890,929	1,043,654,817
VAR 99%	409,167,681	313,528,967	176,157,743	1,003,277,448
VAR 99.5%	398,011,993	306,740,507	169,058,887	984,047,895
ES 95%	422,221,206	323,805,510	182,508,082	1,018,559,469
ES 99%	394,968,716	302,860,015	167,041,738	980,290,063

The simulations reveal that the expected annual return is Đ107M. interestingly, it is worth noting that the standard deviation is quite low relative to the mean which is a result of the policy limits and deductibles truncating the distribution and tail risk. Hence, cargo loss insurance still see a positive return the amongst the worst 99.5% of the simulation losses. Long term results similarly show a favourable outlook, with an expected PV of return at Đ1.143 billion over the total of the next 10 years.

Scenario testing

2 sets of stress testing were conducted to see the return outlook generated from a moderate and extreme levels of stress. (see below to understand what constituted moderate and extreme level of stress and distribution tables)

Moderate stress

Loss statistics	Helionis	Bayesia	Oryn	Total
Expected loss	310,672,653	192,942,191	100,142,366	603,757,210
Standard deviation of loss	15,081,319	11,810,197	8,695,871	20,910,590
VAR 95%	335,161,314	212,586,616	114,652,605	637,900,562
VAR 99%	346,740,955	220,087,810	120,857,006	653,603,041
VAR 99.5%	351,249,905	223,720,568	123,298,265	658,531,782

Return statistics	Helionis	Bayesia	Oryn	Total
Expected return	21,332,113	- 1,329,389	3,303,268	23,305,992
Standard deviation of loss	15,081,319	11,810,197	8,695,871	20,910,590
VAR 95%	- 3,156,548	- 20,973,814	- 11,206,971	- 10,837,360
VAR 99%	- 14,736,189	- 28,475,008	- 17,411,372	- 26,539,839
VAR 99.5%	- 19,245,139	- 32,107,766	- 19,852,631	- 31,468,580

The simulations for moderate level of stress demonstrate the expected claim cost rise by approximately 9%, 27% and 23% for the Helionis, Bayesia and Oryn solar systems respectively. Under moderate correlated stress, the overall portfolio had a positive return in 99.5% of the scenarios, suggesting that the portfolio remains profitable for the most part even under moderate stress. Specifically, Helionis solar system demonstrated resilience to the stress, justifying its higher loss ratio whilst Bayesia was most sensitive due to its initially lower risk baseline. Under extreme stress, claim loss rose by 79% resulting in an expected return of -Đ301M. Despite the negative return, this is not too bad given this an unlikely worst-case scenario assuming the risk is correlated and experience by all 3 solar systems. The results highlight that the policy limit and deductible structure work effectively in mitigating risk, with risk mainly being driven from frequency rather than claim size.

Figure 1- derivation of expected number of shipments needed to meet future ore production demands

Solar system	Helionis	Bayesia	Oryn
Current ore production (tons 000's)	375.00	250.00	125.00
Expansion ore production (tons 000's)	468.75	312.50	143.75
Total fleet capacity (tons 000's)	143.55	139.60	95.78
Trips per fleet vessel (current level)	2.61	1.79	1.31
Trips per fleet vessel (expansionary level)	3.27	2.24	1.50

Figure 2- max shipment weight insured

Vessel	Maximum Volume (kgs)
DeepSpace Haul Box	25,000
DockArc Freight Case	50,000
HardSeal Transit Crate	100,000
LongHaul Value Canister	150,000
Quantum Crate Module	250,000

Pricing

The premium for derived was Đ349M for the Helionis system, Đ202M for the Bayesia system and Đ109M for the Oryn delta system, totalling Đ660M. the premium composition begins with the pure premium (expected simulation loss) which was then adjusted to a Buhlmann credibility premium across the different cargo types being insured, blending the portfolio experience with the container type level data to stabilise the estimate. The present value of claims was computed assuming a uniform distribution of claim arrival with stress loading and volatility loading factor reflecting risks associated stress tests and simulation volatility. A cost of capital margin was also added. The assumed cost of capital for GGIC was 7.46% which is the median worldwide WACC median across a sample of insurance companies (Ramlall, 2025) and thus a suitable proxy This cost of capital was applied to the reserves held which based on the solvency 2 regulatory framework, must be at least enough to cover its liabilities at a 99.5% confidence level (VAR0.995). finally, 5% of the total premium is assumed to fund operational expenses. The rationale of the 5% operational expense was based on the assumption that in the future of 2175, technology improvements has increased operational efficiency, thereby lowering operational expenses. Overall, the loss ratio for the solar systems ended up being 82% for the Helionis sector and 75% for the Bayesia and Oryn solar system. This is comparable with most cargo insurance loss ratios with the average loss ratio for cargo insurers in the EU over the last decade from 2025 being 70.2%. The slightly higher loss ratio is reasonable given the higher risk of loss, especially given the lack of shipment portfolio data.

Stress testing

- **Moderate:** The assumptions that were changed to drive moderate levels of stress was assuming a 5% increase in probability that a vessel makes a shipment, thus increasing the expected shipment frequency by 5%. The other variable altered was a more stressed route risk distribution. Instead of assuming a slightly adjusted historical route risk distribution (symmetrically distributed about route risk level 3), we instead stressed the situation to see how a uniform distribution from route risk level 3-5 would impact the Helionis and Oryn Delta system and a slightly more favourable route risk distribution for the Bayesia system (given its superior communication and mapping systems) would impact results.
- **Extreme:** involved assuming shipments experience the maximum route risk (5), solar radiation (1), debris density (1) and a 10% increase in shipment probability,

Extreme stress test-cargo loss results

Loss statistics	Helionis	Bayesia	Oryn	Total
Expected loss	469,953,066	313,192,468	144,932,965	928,078,500
Standard deviation of loss	18,859,094	15,610,638	10,716,114	26,651,848
VAR 95%	501,305,656	339,445,930	162,795,482	972,238,606
VAR 99%	513,814,270	349,006,182	169,999,924	989,935,739
VAR 99.5%	518,312,665	352,960,989	173,142,523	996,124,123

Return statistics	Helionis	Bayesia	Oryn	Total
Expected return	- 137,948,300	- 121,579,666	- 41,487,331	- 301,015,298
Standard deviation of loss	18,859,094	15,610,638	10,716,114	26,651,848
VAR 95%	- 169,300,890	- 147,833,128	- 59,349,848	- 345,175,404
VAR 99%	- 181,809,504	- 157,393,380	- 66,554,290	- 362,872,537
VAR 99.5%	- 186,307,899	- 161,348,187	- 69,696,889	- 369,060,921

Singular severity GLM model (interpretation model)

```
Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)            1.161e+01  5.751e-02 201.893  < 2e-16 ***
cargo_typegold         5.255e+00  5.210e-02 100.876  < 2e-16 ***
cargo_typelithium      5.008e-01  2.968e-02  16.872  < 2e-16 ***
cargo_typeplatinum     3.866e+00  5.246e-02  73.706  < 2e-16 ***
cargo_typerare earths -1.995e-02  4.379e-02  -0.455    0.649
cargo_typesupplies    -1.775e+00  3.528e-02 -50.318  < 2e-16 ***
cargo_typetitanium    -2.051e+00  2.918e-02 -70.288  < 2e-16 ***
weight                 7.531e-06  3.027e-07  24.884  < 2e-16 ***
route_risk2            2.502e-01  2.979e-02   8.399  < 2e-16 ***
route_risk3            4.605e-01  2.751e-02  16.735  < 2e-16 ***
route_risk4            5.667e-01  2.999e-02  18.896  < 2e-16 ***
route_risk5            7.514e-01  3.415e-02  22.003  < 2e-16 ***
solar_radiation        5.019e-01  6.615e-02   7.587 3.36e-14 ***
debris_density         3.232e-01  6.998e-02   4.619 3.88e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Claim frequency GLM model

```
Coefficients:
                                     Estimate Std. Error z value Pr(>|z|)
(Intercept)                        -0.8228447  0.0325663 -25.267  < 2e-16 ***
cargo_typegold                     -0.0055542  0.0221145  -0.251  0.80169
cargo_typelithium                   0.0179696  0.0220248   0.816  0.41457
cargo_typeplatinum                  0.0243536  0.0219758   1.108  0.26777
cargo_typerare earths              -0.0195750  0.0221581  -0.883  0.37701
cargo_typesupplies                  0.0205921  0.0218953   0.940  0.34697
cargo_typetitanium                  0.0283600  0.0219122   1.294  0.19558
route_risk2                         0.1190416  0.0241989   4.919 8.69e-07 ***
route_risk3                         0.1788803  0.0221480   8.077 6.66e-16 ***
route_risk4                         0.2392424  0.0238266  10.041  < 2e-16 ***
route_risk5                         0.2611979  0.0268885   9.714  < 2e-16 ***
pilot_experience                   -0.0060130  0.0011772  -5.108 3.26e-07 ***
container_typeDockArc Freight Case -0.0086449  0.0187403  -0.461  0.64458
container_typeHardSeal Transit Crate  0.0574221  0.0184563   3.111  0.00186 **
container_typeLongHaul Vault Canister -0.0002024  0.0187146  -0.011  0.99137
container_typeQuantumCrate Module   0.0370491  0.0185514   1.997  0.04581 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

GLM classifier model (2 layered severity model)

```
Coefficients:
                        Estimate Std. Error z value Pr(>|z|)
(Intercept)           -7.090e+00  3.105e-01 -22.835  < 2e-16 ***
cargo_typegold         1.244e+01  5.214e-01  23.853  < 2e-16 ***
cargo_typelithium      1.165e+00  1.137e-01  10.249  < 2e-16 ***
cargo_typeplatinum     8.561e+00  2.777e-01  30.833  < 2e-16 ***
cargo_typerare earths -3.263e-01  2.969e-01  -1.099    0.272
cargo_typesupplies    -4.208e+00  1.010e+00  -4.166 3.09e-05 ***
cargo_typetitanium    -6.308e+00  1.004e+00  -6.284 3.30e-10 ***
weight                 1.717e-05  1.568e-06  10.948  < 2e-16 ***
route_risk2            6.994e-01  1.566e-01   4.466 7.98e-06 ***
route_risk3            1.156e+00  1.452e-01   7.959 1.74e-15 ***
route_risk4            1.489e+00  1.561e-01   9.537  < 2e-16 ***
route_risk5            1.946e+00  1.673e-01  11.632  < 2e-16 ***
solar_radiation        1.378e+00  3.207e-01   4.297 1.73e-05 ***
```

GLM small claim size model (2 layered severity model)

```
Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)            1.165e+01  4.260e-02 273.582  < 2e-16 ***
cargo_typegold         2.152e+00  3.851e-01   5.587 2.34e-08 ***
cargo_typelithium      3.608e-01  2.143e-02  16.839  < 2e-16 ***
cargo_typeplatinum     1.986e+00  6.573e-02  30.211  < 2e-16 ***
cargo_typerare earths  2.437e-02  3.088e-02   0.789     0.43
cargo_typesupplies    -1.666e+00  2.486e-02 -67.005  < 2e-16 ***
cargo_typetitanium    -1.849e+00  2.072e-02 -89.207  < 2e-16 ***
weight                 6.661e-06  2.144e-07  31.071  < 2e-16 ***
route_risk2            2.286e-01  2.454e-02   9.318  < 2e-16 ***
route_risk3            4.021e-01  2.272e-02  17.698  < 2e-16 ***
route_risk4            5.153e-01  2.489e-02  20.702  < 2e-16 ***
route_risk5            6.342e-01  2.836e-02  22.363  < 2e-16 ***
solar_radiation        3.903e-01  5.496e-02   7.102 1.27e-12 ***
debris_density         2.681e-01  5.852e-02   4.581 4.66e-06 ***
---
```

GLM large claim size model (2 layered severity model)

```
Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)           14.25996    0.07687 185.497  < 2e-16 ***
cargo_typegold         2.76507    0.06563  42.131  < 2e-16 ***
cargo_typelithium      0.15804    0.08205   1.926   0.0541 .
cargo_typeplatinum     1.41391    0.06572  21.514  < 2e-16 ***
cargo_typerare earths -0.18230    0.21446  -0.850   0.3953
cargo_typesupplies    -0.05879    0.98364  -0.060   0.9523
cargo_typetitanium     0.93542    0.98373   0.951   0.3417
route_risk2            0.20831    0.04450   4.681 2.90e-06 ***
route_risk3            0.43471    0.04075  10.668  < 2e-16 ***
route_risk4            0.51023    0.04356  11.714  < 2e-16 ***
route_risk5            0.67913    0.04945  13.733  < 2e-16 ***
solar_radiation        0.46196    0.09554   4.835 1.35e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Workers Compensation

Premium Formula: The gross pure premium per worker is the product of a base pure premium and six sequential multiplicative loadings.

Loading Derivations:

1. Trend Factor: 1.2302

The trend factor adjusts historical pure premiums to the policy year 2175 price level. The 5-year average annual inflation rate from the Tarrodan economic dataset is 4.23%, giving a compounded 5-year trend of $(1.0423)^5 = 1.2302$. Inflation affects both the frequency component (through wage growth increasing the salary-linked disability benefit) and the severity component (through medical cost inflation). A single composite trend factor is applied rather than separate frequency and severity trends because the underlying drivers are common to both.

2. IBNR Reserve: 1.12

Workers' compensation is a long-tail line meaning claims are incurred throughout the policy year but many are not reported or fully developed by year-end. The 1.12 IBNR factor reserves for 12% of ultimate losses above the reported amount. This is consistent with NCCI benchmark IBNR factors for high-hazard WC environments (typically 8–15% for frontier and remote operations) and reflects the additional reporting lag at Bayesia and Oryn Delta from communication delays. The factor is applied uniformly across all three systems as a conservative estimate pending the development of system-specific IBNR triangles from operational data.

3. Commercial Loading: 1.6667 (targeting 60% loss ratio)

The gross-up factor converts the net pure premium (expected losses) into a gross premium that covers all operating costs and a target profit margin. The factor is derived from the target loss ratio of 60%: gross-up = 1 / 0.60 = 1.6667. The 60% loss ratio implies 40% available for expenses (22% for acquisition, administration, and general expenses, 12% for loss adjustment expenses and 6% profit load). This is calibrated against NCCI combined ratio benchmarks for commercial WC with the industry benchmark combined ratio of approximately 86% (at a 10% investment return assumption) implies a 14% underwriting profit, consistent with our 6% explicit profit load plus the investment income generated on the premium float.

4. Catastrophe Provision: 1.05

A 5% catastrophe loading is applied above the commercial gross-up. The NCCI cat provision for standard WC is 0.5–2.5% with the elevated 5% chosen reflecting the Year 1 frontier surcharge with three main motivating reasons:

- No operational claims data from Bayesia or Oryn Delta to validate the modelled tail
- The potential for correlated multi-system events from galaxy-wide solar storms
- The genuine probability of unmodelled hazards in frontier environments that do not appear in historical data. This provision will be reviewed at each renewal and reduced as operational experience accumulates.

5. Investment Income Offset and Combined Ratio

Workers' compensation claims are long-tail as mentioned above and hence the assumed average lag between premium collection and final claim settlement is approximately three years. This means Galaxy General can earn investment income on the held premium reserve during this period where at our base 10% investment assumption, investment income per policy year is GP × 0.10 = Ð1.701M for the portfolio.

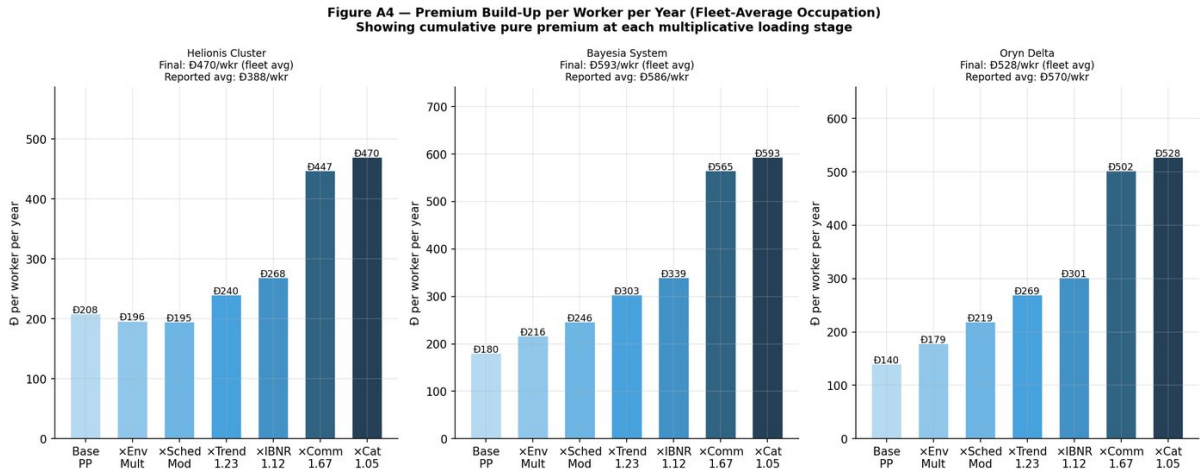
6. Schedule Modifcation Factor

The schedule modification factor adjusts the fleet-average pure premium for CQ's specific workforce risk profile relative to the historical pool from which rates were derived. It is computed using the five GLM coefficients (excluding gravity, which is captured in M_env):

$$M_sched = \exp(\Sigma_j \beta_j \times (CQ_j - pool_j))$$

Where:

- β_j = GLM coefficient for predictor j | CQ_j = CQ workforce average for predictor j
- pool_j = historical pool average for predictor j | j ∈ {experience, psych_stress, safety_training, accident_history}



This figure shows the cumulative premium per worker at each loading stage for all three systems. The gap between Helionis (Ð388 reported average) and Bayesia (Ð586) opens at the environment and schedule modification stages, confirming that the premium differential is driven by the EM radiation residual loading and Bayesia's younger, more stressed frontier workforce rather than by base claim cost differences. Oryn Delta's final premium of Ð570 despite its lower base pure premium (Ð140 vs Ð180 for Bayesia) reflects its larger combined environment and schedule modification debit.

Buhlmann Credibility Calculations

Pure premiums differ by occupation class because workers in different roles face materially different physical hazards, management quality, and operational exposure. Bühlmann credibility blends the class-specific observed rate with the fleet-wide prior in proportion to the volume of data available for that class. The credibility weight Z determines the blend:

$PP_class = Z \times PP_observed + (1 - Z) \times PP_fleet$
Where:

- $Z = E / (E + k)$ where $k = EPV / VHM$
- E = exposure in worker-years for the class
- EPV = Expected Value of Process Variance
- VHM = Variance of the Hypothetical Means across classes

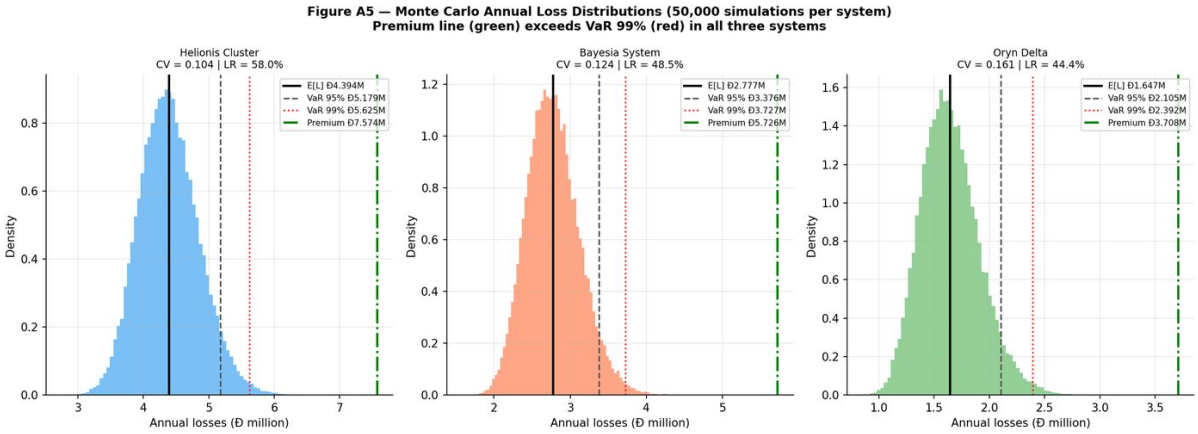
The key parameter k = EPV/VHM = 254.2 worker-years represents the exposure at which Z = 0.50: a class with exactly 254 worker-years receives equal weight from its own data and the fleet prior. Classes with more exposure receive more weight on their own experience and classes with less data are pulled toward the fleet average. This prevents any single class, particularly those with few observations like Executives from being priced on a handful of unrepresentative claims.

Occupation class	Exposure (wy)	Z	Freq rel.	Mean sev (Đ)	Sev rel.	PP rel.
Drill Operator	13,394	0.981	**1.668×**	7,040	**0.900×**	**1.501×**
Executive	681	0.728	**0.649×**	17,531	**2.241×**	**1.453×**
Engineer	9,911	0.975	**1.054×**	9,677	**1.237×**	**1.304×**
Technology Officer	3,368	0.929	**1.088×**	6,921	**0.885×**	**0.963×**
Scientist	3,359	0.928	**0.868×**	8,613	**1.101×**	**0.956×**
Maintenance Staff	2,745	0.914	**1.110×**	6,003	**0.767×**	**0.851×**
Safety Officer	6,772	0.963	**0.860×**	7,614	**0.973×**	**0.837×**
Planetary Ops	6,772	0.963	**0.818×**	7,059	**0.902×**	**0.738×**
Spacecraft Op.	3,394	0.928	**0.784×**	6,132	**0.784×**	**0.615×**
Administrator	6,883	0.964	**0.568×**	5,697	**0.728×**	**0.413×**

Capital Modelling
Aggregate annual losses are simulated by 50,000 iterations per system. Each iteration proceeds as follows

- Draw annual claim count N from Poisson(λ), where λ is the occupation-weighted fleet-level expected claim countl
- For each of the N claims, draw an individual claim amount X from LogNormal(μ_adj, σ), where μ_adj incorporates the occupation-weighted severity relativity, environment severity multiplier, trend, and IBNR adjustments

- Compute aggregate annual loss S = ΣX for the iteration. The resulting empirical distribution of S is used to estimate VaR, TVaR, and net revenue under each scenario.



The figure above shows the simulated annual loss distributions for all three systems First, all three distributions have the annual gross premium (green line) above the VaR 99% (red dotted line), confirming that Galaxy General retains a positive underwriting margin even in the worst expected annual event at every individual system. Second, the distributions are approximately LogNormal in shape which is consistent with the aggregate being the convolution of a Poisson number of LogNormal amounts. Third, the coefficient of variation increases from Helionis (0.104) to Bayesia (0.124) to Oryn Delta (0.161), reflecting the decreasing workforce size and increasing parameter uncertainty as we move from the most to least data-rich system.

Full-Per System Statistics

Statistic	Helionis	Bayesia	Oryn Delta	Portfolio[1]
Annual gross premium	Đ7.574M	Đ5.726M	Đ3.708M	Đ17.008M
E[L] — expected annual loss	Đ4.394M	Đ2.777M	Đ1.647M	Đ8.818M
σ[L] — std deviation	Đ0.455M	Đ0.345M	Đ0.265M	Đ0.786M
CV = σ/E[L]	0.1035	0.1243	0.1610	0.089
VaR 90%	Đ4.980M	Đ3.218M	Đ1.982M	—
VaR 95%	Đ5.179M	Đ3.376M	Đ2.105M	—
VaR 99%	Đ5.625M	Đ3.727M	Đ2.392M	Đ11.74M[2]
VaR 99.5%	Đ5.828M	Đ3.881M	Đ2.540M	—
TVaR 99%	Đ5.953M	Đ3.967M	Đ2.650M	Đ12.57M[2]
Loss ratio	58.0%	48.5%	44.4%	51.8%
Combined ratio	92.0%	82.5%	78.4%	85.9%
Net revenue (10% inv.)	Đ0.983M	Đ1.289M	Đ0.986M	Đ3.258M

Net revenue (5.1% risk free rate)	Đ0.612M	Đ1.008M	Đ0.805M	Đ2.425M
Investment income (10%)	Đ0.757M	Đ0.573M	Đ0.371M	Đ1.701M
Investment income (5.1%)	Đ0.386M	Đ0.292M	Đ0.189M	Đ0.867M

Business interruption: this is done based on Option 1

The zero-inflated negative binomial (ZINB) frequency model utilises a constant zero-inflation parameter (zi=1), which yielded the highest AIC/BIC performance amongst variable zero-inflation parameter model and models with interaction terms. This suggests that a structural zero state where no claims occur, is likely a universal behaviour of autonomous extraction fleets and self-healing hulls used across all systems. Roughly 88% of the portfolio resides in this zero-claim state, independent of solar system hazards.

A Monte-Carlo simulation with 20000 iterations was conducted using frequency sampled from ZINB model and severity sampled from Gamma model (shape parameter 1.41, scale parameter 1085000). Simulated individual claims are strictly bounded between 28000 deductible and 1426000 policy limit. Based on a 22% commercial loading and Bulmann framework, per-unit premium for each system is projected.

System	Per-unit premium (credibility factor 0.1914)
Safe	$16.28
Average	$20.14
Catastrophic	$19.81

A system of zero claim, average claim frequency and severity, and a catastrophic system with highest possible claim amount were simulated. The Catastrophic premium is slightly lower than the Average premium, as the upper threshold limit essentially censors the loss, making a single catastrophic hit statistically less "expensive" to the model than a system that might have multiple smaller, high-frequency claims that sum to a higher theoretical total.

A.4 Risk Assessment

RCD Tool

Risk Category	Risk Subcategory	Risk Division	Risk	Risk Definition

Operational Risk	Mechanical / asset reliability	Equipment Failure	Aging machinery	Failure risk arising from older mining equipment with higher susceptibility to breakdown.
Operational Risk	Mechanical / asset reliability	Equipment Failure	Maintenance interval risk	Failure risk caused by infrequent maintenance cycles.
Operational Risk	Mechanical / asset reliability	Equipment Failure	High utilization stress	Increased breakdowns from equipment overuse.
Operational Risk	Infrastructure resilience	Business Interruption	Power system inadequacy	Interruption risk from insufficient backup systems to sustain activity during outages.
Operational Risk	Infrastructure resilience	Business Interruption	Production concentration risk	Greater interruption severity where operations depend heavily on a small number of processes.
Operational Risk	Infrastructure resilience	Business Interruption	Safety-compliance disruption risk	Business interruption caused by investigations/mandated shutdowns following poor compliance outcomes.
Supply Chain Risk	Input and logistics dependency	Business Interruption	Input sourcing fragility	Interruption risk caused by reliance on limited or unstable supply channels for materials.
Supply Chain Risk	Input and logistics dependency	Cargo Loss	Route disruption risk	Cargo exposure created by hazardous routes subject to debris, detours, congestion, etc.
Supply Chain Risk	Input and logistics dependency	Cargo Loss	Container/packaging failure	Risk from the containment system being unsuitable for the cargo type or journey.
Transport Risk	Transit environment	Cargo Loss	Solar radiation exposure	Cargo degradation caused by radiation conditions during transport.
Transport Risk	Transit environment	Cargo Loss	Long-duration transit risk	Longer travel times prolonging exposure to environmental and operational hazards.
Transport Risk	Vessel and crew	Cargo Loss	Pilot experience risk	Increased transport loss exposure associated with less experienced pilots making errors.
Transport Risk	Vessel and crew	Cargo Loss	Aging vessel risk	Cargo exposure linked to vessels with outdated systems or higher maintenance needs.
Transport Risk	Cargo profile	Cargo Loss	High-value cargo concentration	Severity risk when individual shipments contain very high insured values.
Workforce Risk	Worker safety and injury	Workers' Compensation	Training deficiency/inexperience	Injury caused by inexperience or insufficient safety training in hazardous environments.
Workforce Risk	Worker safety and injury	Workers' Compensation	Fatigue/hours-worked risk	Risk from long working hours, fatigue, and reduced alertness in demanding roles.
Workforce Risk	Worker safety and injury	Workers' Compensation	Protective gear insufficiency	Injury caused by poor-quality or unsuitable protective equipment.
Workforce Risk	Worker safety and injury	Workers' Compensation	Prior accident history risk	Higher claim likelihood associated with workers with prior incidents.
Workforce Risk	Worker wellbeing	Workers' Compensation	Psychological stress injury	Stress-related claim exposure arising from demanding conditions, isolation, pressure, or traumatic events.
Workforce Risk	Environment-specific human exposure	Workers' Compensation	Gravity-related injury risk	Injury risk from working in gravity conditions different from Earth, affecting movement, lifting, and equipment use.
Environmental / Space Hazard Risk	External environmental shocks	Cross-hazard	Solar storm/debris/asteroid type event	An external event that may damage equipment, interrupt operations, endanger workers, and affect cargo transit.
Concentration Risk	Portfolio structure	Cross-hazard	Single-client concentration	Risk that dependence on one mining client creates earnings volatility and accumulation exposure.

Dependency / Correlation Risk	Multi-line dependency	Cross-hazard	Cascading loss risk	A single initiating event triggers losses across other hazard areas, such as equipment failure causing business interruption and worker injury.
Financial Risk	Cost and pricing environment	Cross-hazard	Inflation risk	Future costs rise faster than expected, eroding margins and increasing reserve and pricing uncertainty.
Financial Risk	Cost and pricing environment	Cross-hazard	Interest rate risk	Present values of cash flows change as term rates and discount assumptions move over time.
Strategic / Product Risk	Coverage design	Cross-hazard	Underinsurance/underwriting risk	Coverage structure does not properly match actual values at risk or loss scenarios.
Data / Model Risk	Information quality	Cross-hazard	Incomplete historical relevance	Historical claims data may not fully reflect future conditions.
Data / Model Risk	Information quality	Cross-hazard	Data inconsistency risk	Inconsistent naming or categorisation across files may distort assumptions and analysis.
Emerging / Future Risk	Automation and robotics	Equipment Failure / Workers' Compensation	Human-machine interface risk	Increased automation creates new exposures from malfunction and worker interaction with machinery.
Emerging / Future Risk	Expansion and scalability	Cross-hazard	Expansion risk	Entry into new routes or solar systems creates exposure profiles outside the original underwriting envelope.

Risk Index Table

Equipment Type	Helionis Cluster	Bayesia System	Oryn Delta	Mean Risk
Quantum Bores	0.69	0.77	0.93	0.80
Graviton Extractors	0.48	0.57	0.73	0.59
Fexstram Carriers	0.67	0.71	0.78	0.72
ReglAggregators	0.24	0.26	0.35	0.28
Flux Riders	0.24	0.21	0.20	0.22
Ion Pulverizers	0.64	0.66	0.75	0.68
Mean	**0.49**	**0.53**	**0.62**	

Rank	Threat	Systems Affected	Likelihood	Severity	Mitigation
1	Catastrophic joint solar event	All	0.01	Extreme	Shock modelling in monte carlo
2	Oryn Delta flare events	Oryn Delta	Medium	High	High deductible and credibility factor

3	Quantum bores and ion pulverizers failures	All	Medium	High	Risk accounted for by GLM
4	Bayesia System radiation spike (orbital alignment)	Bayesia System	Medium	Moderate	Radiation exclusion clause; Z = 0.60 pricing buffer
5	Helionis asteroid cluster fragmentation	Helionis Cluster	Medium	Moderate	Communications relay resilience; maintenance scheduling
6	Equipment age-related frequency increase	Helionis Cluster	High	Moderate	Age coefficient in GLM and annual rate review at renewal
7	Model uncertainty in proxy systems	Bayesia, Oryn Delta	High	Moderate	Bühlmann credibility weighting
8	Fleet expansion outpacing claims data	All	Medium	Low	Annual repricing to be conducted over time

Rank	Threat	Systems Affected	Likelihood	Severity	Mitigation
1	Catastrophic joint solar event	All	1%	Extreme	Shock modelling in monte carlo
2	Oryn Delta flare events	Oryn Delta	Medium	High	High deductible and credibility factor
3	Quantum bores and ion pulverizer failures	All	Medium	High	Risk accounted for by GLM
4	Bayesia System radiation spike (orbital alignment)	Bayesia System	Medium	Moderate	Radiation exclusion clause; Z = 0.60 pricing buffer
5	Helionis asteroid cluster fragmentation	Helionis Cluster	Medium	Moderate	Communications relay resilience; maintenance scheduling
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Cargo loss

Threat table

- One of the top threats to cargo loss is high shipment frequency. Since we're insuring a fleet with uncertain volume of shipments, greater shipment occurrence compound the total loss impact. This risk was hedged through policy design of capping shipments. However, fully utilising all shipments would create risk given modelling assumed not all shipments would be needed to meet expansionary mining goals
- Another top threat to cargo loss is that the route risks, solar radiation and debris density (covariates increasing claim frequency and severity) of the 3 solar systems do not reflect historical data distributions. Whilst this was hedged by adjusting historical risk probabilities based on the qualitative description of the 3 Solar systems, an underestimation of the adjustments could result in our premium being under-priced
- A key threat to cargo loss was the insured only insures high valuable cargo types. The exposure was minimised through a policy limit scheme as well as exclusion and weight restriction of gold and platinum respectively

Narrative of solar system risk differences

- Bayesia offers the greatest stability for cargo loss, having reliable communication systems and standardised supply routes. It does suffer however form ambient radiation levels and occasional electromagnetic spike
- Helionis has moderately high level of risk. Although low solar activity and predictable winds, it suffers from irregular gravitation resonance leading to high debris density and micro collisions that can damage cargo
- Oryn delta presents a great level of risk amongst the 3 solar systems. The asymmetric asteroid ring creates zones of rapidly fluctuating gravitational gradient compounded by low light conditions poses significant risk to the cargo journey.

A.5 Assumptions

Equipment Failure

- The distribution of claim frequency and severity remains constant throughout the years for long-term models
- Growth of the equipment count is constant and flat
- Constant inflation and risk-free rate based on the average of previous years
- 1-in-100-year events impact solar systems through the year, these are catastrophic years
- The shock factor and probability of galaxy wide shocks remains constant
- The magnitude of the shock was assumed to multiply the negative binomial size parameter by 5, increasing the number of claims made in the year
- The Bühlmann credibility weighting remains constant (No way to change without incoming data)

Cargo loss

- The shipment frequency follows a binomial distribution with parameters chosen to be in line with the maximum number of shipments allowed by the policy and ensure the expected shipments is equal to the number needed to meet each of the solar systems long-term mining production goals assuming shipments are made at full capacity
- Historical covariate distributions is representative of future shipments in the Helionis, Bayesia and oryn delta system. Small quantitative adjustments were also made to the covariate sampling based on qualitative description
- Policy constraints are intentionally adhered to. For example, if CSMQ is to make a platinum cargo shipment, they intentionally choose a weight less than 50000kg to allow their shipment to be insured
- Assumed that claim frequency and claim severity of shipments were independent. The rationale being such a structure is typically used in insurance
- Inflation rate was modelled using a normal distribution, assuming that inflation rate is anchored to a target value with the government adjusting interest rates to combat deviations. The dataset shows a strong correlation between interest and inflation rates with high inflation rates being associated high interest rates. Thus, we modelled real interest rates using an auto-regressive model. By modelling an interest and inflation rate, we developed a link between interest and inflation

Workers Compensation

- Claim frequency is modelled using a Poisson distribution (MLE), supported by an overdispersion index of 0.996–1.020 and Negative Binomial $\theta > 100$, indicating minimal overdispersion The Poisson assumption will be reviewed if the overdispersion index exceeds 1.10 at renewal
- Claim severity is modelled using a LogNormal distribution (MLE), with a strong AIC margin (218–467) over the Gamma distribution across all systems and this assumption will be reviewed if new data materially changes the AIC comparison.
- The Bühlmann credibility parameter (k) is set at 254.2 worker-years, derived from EPV = 0.002845 and VHM = 0.0000112 and this will be reviewed annually as additional exposure data accumulates.
- Bayesia's surface gravity is assumed to be 1.350g based on encyclopaedia classification as a high-gravity planet, despite no exact value being providedA pre-binding gravitational survey is recommended at first renewal
- Oryn Delta's surface gravity is assumed to be 1.050g, based on a conservative near-Earth approximation for a rocky exoplanet with no defined descriptor. A gravitational survey is recommended at first renewal.
- Bayesia includes a +10% residual frequency loading (1.10×) to reflect electromagnetic radiation exposure and extreme environmental conditions, with no historical proxy available. This will be reviewed at first renewal using Year 1 claims data

- Bayesia includes a +10% residual severity loading (1.10×), based on the same rationale as the frequency loading. This will be reviewed at first renewal
- Oryn Delta includes a +15% residual frequency loading (1.15×), driven by low visibility, asteroid ring operations, tectonic terrain, and communication delays. This will be reviewed at first renewal
- Oryn Delta includes a +5% residual severity loading (1.05×), reflecting that severity is less sensitive to communication delays than frequency. This will be reviewed at first renewal
- A trend factor of 1.2302 (4.23% per annum) is applied, based on a 5-year average inflation rate from the Tarrodan dataset. This will be reviewed annually
- An IBNR factor of 1.12 (12%) is applied, consistent with long-tail workers' compensation business and accounting for reporting delays in Bayesia and Oryn Delta This will be reviewed annually as development data becomes available
- A catastrophe provision of 1.05 (5%) is included as a frontier risk surcharge, noting that the National Council on Compensation Insurance typically suggests a range of 0.5–2.5% This will be reviewed annually and is expected to reduce as experience matures.
- The base investment return is assumed to be 10% nominal (~5.1% real), consistent with historical long-run equity premium evidence. This will be reviewed annually
- Inter-system correlation is assumed to be $\rho = 0.30$ using a Gaussian copula, reflecting moderate dependence from shared galaxy-wide events This will be reassessed with operational data, with sensitivity tested between 0.10–0.50.
- A special benefit of 3× salary is included for system-specific hazards, covering radiation illness (Bayesia) and asteroid ring exposure (Oryn Delta), reflecting both frequency and severity impacts of novel risks. This will be reviewed at first renewal once claims data emerges
- A Đ250,000 permanent disability floor is applied across all systems to correct inequities in salary-linked benefit structures and ensure a meaningful minimum payout. This will be reviewed if minimum wage regulations change.
- Workforce exposure is proxied using mine count proportions (30/15/10), as only mine-level data is available. This assumption will be updated if CQ provides system-level workforce headcount data

Business Interruption:
As claim amount is adjusted to upper and lower bounds. It is then strictly assumed that claim cannot be made below 28000 or above 1426000. In the case where claims made above or below these thresholds, an alternative is to add a product design feature with a deductible and policy limit set at those bounds. Entering a non-proportional reinsurance contract with a similar excess of loss, however, may not be appropriate, as claim experience is predominantly higher than this threshold.

The model's validity hinges on the stationarity of the zero-inflation parameter (zi=1) during the projected 25% fleet expansion, assuming the 88% "structural zero" reliability

remains a technological constant despite potential orbital saturation or logistical complexities. Furthermore, the analysis assumes a 1-to-1 linear relationship between the number of units and total risk; if systemic interdependencies cause risk to compound non-linearly, the current 22% commercial loading will be insufficient. Finally, the predictive integrity of Energy Backup Scores and Maintenance Frequency is assumed to be absolute, though undetected hardware fatigue in the corrosive Epsilon or Zeta environments could neutralise these proxies, potentially exhaust the $16.73 Billion (VaR 99.5) capital buffer and underestimating the $16.09 Billion expected aggregate loss.

A.6 Limitations

In all datasets, extensive cleaning needed to be conducted in order to utilise the data effectively. Common issues included strings and numbers not being entered correctly with trailing characters and values representing positive variables such as time, being negative. The data was also adjusted to follow the provided ranges, in some cases, these ranges needed to be modified. Another common issue is the conflicting information between the data and the datasheet which suggested different names or had extra names such as in the case with equipment type where extra equipment was listed in the datasheet.

Equipment Failure:

For the purposes of modelling, The Oryn Delta and Bayesian Solar systems have been mapped to the Zeta and Epsilon systems respectively as there is not provided historical data, the rationale behind this is due to the similar planetary conditions. To accommodate for this, Bühlmann credibility weighting was used, where Oryn Delta was given a Z weighting of 0.5, Bayesian System was set to 0.6 and Helionis Cluster to 1.
Station ID’s could not be included in the models as it wasn’t provided in the data
Many of the ranges provided in the datasheet deviated significantly from what the data proposed. For example, claim amounts were given a range, however many claims exceeded the upper bound or were below the lower bound, in this case, the lower and upper bound were uncapped, and any values that were below 0 or were unreasonably low were removed. The highest values could have potentially been catastrophic losses which are important to capture in the model. Equipment age needed to have its range extended from 0-10 to 0-20, since 2599 of the equipment across all solar systems exceed 10 years of age.

Cargo loss
a) external sites listed below in the bibliography were used as a comparison tool to build in assumptions with pricing (such as determining an appropriate cost of capital given the absence of this information) and appropriate loss ratios to corroborate the validity of our pricing scheme.

b)

One key data limitation was a lack of shipment portfolio data. We were simply provided with the number of vessels to be insured and their corresponding maximum cargo weight. Thus, we restricted the number of shipments a vessel can make for each solar system and used a binomial distribution to simulate the number of shipments with the expected value being equal to the expected number of shipments at full capacity that would be needed to reach their ore production levels. The impact of this limitation is that the expected number of shipments needed may be underestimated if CSMQ doesn't typically load their cargo vessels at max weight, requiring more shipments to meet their long-term total mining output goals Another data limitation was the lack of quantitative statistics regarding the solar systems. Only qualitative information was provided where we used judgment to adjust the values of historical data which may however been understated. For example, for route risk simulation for the Oryn system, we increased the probability of route risk 4 and 5 by 10% and 15% respectively based on their fluctuating gravitation gradients. However, this increased may be too conservative and thus underestimate the solar system's risk. This is partially mitigated however with stress loading incorporated in the premium.

Worker's Compensation

The analysis for worker's compensationis subject to several data and modelling limitations that introduce uncertainty into the pricing framework, although each has been addressed with targeted mitigations. First, the absence of historical claims data for Bayesia and Oryn Delta means pricing relies on proxy systems with environmental adjustments, increasing parameter uncertainty and this is mitigated through analogue system mapping, judgemental residual loadings, and an annual premium review rider, with Bühlmann credibility progressively shifting weight toward system-specific experience as data emerges. Second the CQ workforce count is only provided at an aggregate level (35,807), so system-level exposure is inferred using mine count proportions (30/15/10) and while this is a reasonable proxy, deviations would affect the allocation of premium across systems rather than the total portfolio premium. Thirdly, the severity dataset does not include duration or incapacity period information, preventing direct modelling of long-tail worker's compensation benefit phases and instead requiring reliance on aggregate severity. This is mitigated through the application of a 1.12 IBNR factor to capture unreported and underdeveloped claims. Foruth, although GLM predictors are specified at the individual level, they cannot always be cleanly aggregated to the system level, requiring estimation of system-level workforce characteristics from the personnel file. This is addressed by using occupation composition data and computing average risk factors from personnel records. Fifth, Bayesia's assumed gravity of 1.350g is based on encyclopaedic sources rather than direct measurement, meaning environmental loadings may be miscalibrated if actual gravity differs. The sensitivity analysis indicates that a ±0.15g variation changes net result by approximately ±Đ0.26M, implying moderate risk, and a survey is recommended at renewal. Finally, there is no historical data for radiation illness claims at Bayesia, so the benefit is priced structurally (three times salary) rather than through frequency calibration. This is mitigated through conservative multipliers and a 30-day latency provision.

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- Utilised AI to refactor code and improve modularity
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